

# Dynamic modelling and stability parametric analysis of a flexible spacecraft with fuel slosh

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## ABSTRACT

Modern spacecraft often contain large quantities of liquid fuel to execute station keeping and attitude manoeuvres for space missions. In general the combined liquid-structure system is very difficult to model, and the analyses are based on some assumed simplifications. A realistic representation of the liquid dynamics inside closed containers can be approximated by an equivalent mechanical system. This technique can be considered a very useful mathematical tool for solving the complete dynamics problem of a space-system containing liquid. Thus they are particularly useful when designing a control system or to study the stability margins of the coupled dynamics. The commonly used equivalent mechanical models are the mass-spring models and the pendulum models. As far as the spacecraft modelling is concerned they are usually considered rigid; i.e. no flexible appendages such as solar arrays or antennas are considered when dealing with the interaction of the attitude dynamics with the fuel slosh. In the present work the interactions among the fuel slosh, the attitude dynamics and the flexible appendages of a spacecraft are first studied via a classical multi-body approach. In particular the equations of attitude and orbit motion are first derived for the partially liquid-filled flexible spacecraft undergoing fuel slosh; then several parametric analyses will be performed to study the stability conditions of the system during some assigned manoeuvres. The present study is propaedeutic for the synthesis of advanced attitude and/or station keeping control techniques able to minimize and/or reduce an undesired excitation of the satellite flexible appendages and of the fuel sloshing mass.

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## 1. Introduction

The analysis and the study of a sloshing fluid in a solid container has received considerable attention for long time [1–3]. Sloshing is a wave phenomenon, and is driven by inertial forces, surface tension, and surface kinematics. To take all the detailed dynamics into account in a vehicle dynamic modelling is computational time consuming. On the other hand, a complete dynamic-stability analysis of the vehicle must include the forces produced by the liquid motion in partially filled propellant tanks. This is, of course, mandatory to study the mutual interaction between the dynamics of the spacecraft and of its internal fluid. Some detailed studies on the motion of orbiting spacecraft with fuel sloshing can be found in [4] where the general equations of a sloshing fluid in a rigid container are presented for Newtonian potential in orbital (i.e. zero-g or quasi zero-g) conditions. In that research some useful information on the specific effects of the gravitational torque and of sloshing on the overall stability were

provided. In general the combined liquid-structure system is very difficult to model; on account of this the analyses are based on some assumed simplifications. A realistic, even if simplified, representation of the liquid dynamics inside closed containers can be modelled by an equivalent mechanical system. The technique of equivalent mechanical models can be considered a very useful mathematical tool for solving the complete dynamics problem of a system containing liquid. They are also particularly useful when designing a control system or to study the stability margins of the coupled dynamics [5,6]. Following this approach, the sloshing dynamics can typically be represented either by a mechanical model of mass-spring-damper or by an equivalent pendulum. The simple mass-spring-damper model is able to represent the linear lateral sloshing mode whereas the pendulum models have the advantage that the natural frequency varies when changes in acceleration happen. Furthermore, depending on the container's geometric shape, the pendulum model can represent large slosh motions, and the spherical pendulum models are more often used to emulate nonlinear rotary sloshing [7]. In both cases this equivalent model is then included in the equation of motion of the entire vehicle for Guidance, Navigation and Control analysis (GN&C). In [8] an analysis of the interaction of the attitude motion, the fluid motion and the solar arrays vibration was

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## Nomenclature

$m_s, m_f, m_p$  main platform, fluid and solar panel mass (respectively)  
 $I_{cms}, I_{jf}, I_{jp}$  main platform, fluid and solar panel moment of inertia (respectively)  
 $\vec{d}_{jf}$  and  $\vec{d}_{cmf}$  the position of the pivot of the spherical pendulum and the position of the fuel sloshing mass with respect to the fuel reference frame respectively

$\vec{r}_{cms}$  main platform centre of mass position  
 $\vartheta_s, \vec{\omega}_s$  main platform attitude and angular velocity  
 $\vartheta_f$  fluid mass oscillation  
 $w(\xi)$  solar panel elastic displacement  
 $\eta_k(t)$  solar panel  $k$ th modal amplitude  
 $\phi_k(\xi)$  solar panel  $k$ th modal shape  
 $\alpha_k, \beta_k$  translational and rotational modal participation factors  
 $k_{p\delta}, k_{d\delta}$  proportional and derivative gains

performed by taking the spring-mass equivalent model into account. In [9] the pendulum slosh model is used to study the effects of slosh during a orbit change manoeuvre. Similar approach is presented in [10] where the pendulum model is used to model the fluid-attitude interaction on an accelerating rocket upper stage moving in a single plane to implement a wave-based controller. Finally in [11] the dynamics of a rigid-spacecraft with a spherical fuel tanks by taking the pendulum-model or the mass-spring model into account was described and the problem on the Lyapunov-based controller design was analyzed.

From the control design point of view the quantities that are input to the GN&C analysis to consider the propellant sloshing are: natural frequency of the sloshing, sloshing mass, sloshing mass centre coordinates, and critical damping coefficient or the viscous torsional damping of the fluid [12]. As far as the spacecraft modelling is concerned, they are usually considered rigid; i.e. no flexible appendages such as solar arrays or antennas are considered when dealing with the interaction of the attitude dynamics with the fuel slosh. On account of this no special emphasis has been devoted to the coupling of the liquid motion, the attitude dynamics and the flexibility of the spacecraft.

The aim of this paper is to analyse the interactions among the control, the attitude dynamics, the flexibility of the solar array and the sloshing motion of spacecraft using a multi-body approach. On account of this, firstly a complete formulation of a 3D flexible space-structure with a sloshing mass was described in Section 2. Then a 2D model was used to study and understand the most significant coupling effects appearing when the attitude dynamics, the attitude control and the flexible elements of the spacecraft interact among each other.

The paper is organized as follows. In Section 2 we present the mathematical equations of the liquid filled flexible spacecraft under gravity and gravity gradient forces considered as a multibody system. In Section 3 the simplified 2D flexible satellite equations are reported. These equations will be eventually used in Section 4 to perform some parametric stability analysis when the spacecraft is under the action of the thrust and of the attitude control. Many non-linear analyses will be also presented and discussed in the present section by evaluating the time history response of the flexible structure with the fuel-sloshing mass when some demanding manoeuvres are imposed. Finally in Section 5 the final remarks will be presented.

## 2. Governing equations of a flexible spacecraft with fuel slosh

The mathematical approach we used to model a flexible spacecraft with fuel sloshing is based on a multibody approach where the equations of motion are firstly written for each element (i.e the central bus, the solar array and the fuel spherical pendulum) of the spacecraft system [13]. Then, by considering the Newton Euler formulation [14], the equations of motion are assembled by introducing

the reaction forces and compatibility conditions. In fact, in the Eulerian approach, the motion of each individual member is described through kinematic parameters: (a) position of its Centre of Mass (CM) with respect to the inertial frame; (b) rotation of the members with respect to the inertial frame; (c) amplitude of the elastic modes for the elastic bodies when the well-known modal decomposition technique is adopted. We can underline, however, that the above-described motion parameters are redundant, so that we must introduce: (a) reactions between members, (b) equations of compatibility of the same number of reactions. In summary, (i) the set of unknowns is: motion parameters, reactions, control forces; (ii) the equations are equilibrium, compatibility, control. This leads to a set of Differential Algebraic Equations (DAE), which are demanding to be solved from a computational point of view. To overcome this problem, in the present paper, we adopted a set of parameters, which are already consistent with the compatibility conditions. On account of this no compatibility conditions appear and the unknown reactions can be expressed in terms of the inertial, elastic and applied forces.

### 2.1. Mechanical model of the fluid slosh

In the present work, the mechanical equivalent model of the fuel motion inside the satellite is represented by a pendulum with a mass  $m_f$  located at the end of a weightless bar with a fixed length  $\|\vec{d}_{cmf}\|$  and a fixed part of the fluid which moves with the spacecraft [15]. The fuel sloshing pendulum-like model is often used in literature because it automatically restricts the motion of the fuel. In fact in this case, contrary to the spring-dampers models, the fuel cannot move outside the tank. Without any loss of generality, the fixed fraction of the fuel mass can be incorporated into the mass of the main spacecraft and hence the body fixed spacecraft reference frame is evaluated considering this. Of course, this assumption is valid if the fluid does not disintegrate into fractional masses.

As far as the friction between the fuel and the tank wall is concerned, it is here modelled with a torsional damping. The damping coefficient, of course, will depend on the propellant properties, tank shape and dimensions, and internal baffles (if any). The layout of the three different bodies representing the main spacecraft, the solar panel and the fuel is reported in Fig. 1

### 2.2. Single bodies governing equations

We recall here some general relationships that are useful to derive the governing equation of the multibody spacecraft system. More details can be found in [13]. Suffice here to say that the position  $\vec{r}_p$  of a generic point P of a body (either solid, elastic or liquid), with respect to an Earth-fixed reference frame, can be expressed as:

$$\vec{r}_p = \vec{r}_{cm} + {}^I\mathbf{T}_{b_j}\vec{r}_b \quad (1)$$

where  $\vec{r}_b$  denotes the position of the same element with respect to a body-fixed reference frame,  $\vec{r}_{cm}$  is the value of  $\vec{r}_b$  for the satellite point

geometrically coincident with the undisturbed centre of mass; the matrix  ${}^I\mathbf{T}_{b_j}$  allows to pass from the body to the inertial reference frame. From Eq. (1) it is possible to evaluate the acceleration of the generic point P of the body in terms of the acceleration of its centre of mass (global motion) and of its elastic components (if any).

Actually, by applying the Newtonian approach it is straightforward to derive the translational and rotational equations of motion.

Of course, if we have an elastic body we must introduce the “structure” through a linear operator  $L[w]$  which transforms the small structural displacement  $w$ , referred to the body reference frame, to structural forces acting on the generic point of the body. The assumption of a “small” displacement leads to the possibility to express the elastic displacements  $w$ , as a combination of the eigenmodes of vibration:

$$w(\xi) = \sum_k \eta_k(t) \phi_k(\xi) \quad (2)$$

where  $\xi \equiv (\xi_1, \xi_2, \xi_3)$  are the coordinates of a point P in the undeformed body, referred to the body axes,  $\phi_k(\xi)$ 's are the eigenmodes of vibration and the coefficient  $\eta_k(t)$  must be determined [14].

Let us consider that the multibody system is constituted by three bodies: the Main Platform (MP) that constitutes the rigid part of the Spacecraft, the solar panel (elastic) attached to the MP through a locked hinge and finally a sloshing mass constrained to the MP with a spherical joint. By following the Newtonian approach, we can write the translational equation of motion of the MP as follows:

*Main platform translational equations*

$$m_s \ddot{\vec{r}}_{cms} = \vec{R}_{jf} + \vec{S} + \vec{R}_{jp} \quad (3)$$

where the vector  $\vec{r}_{cms}$  represents the position of the centre of mass of the main platform with respect to the Inertial Reference Frame (IRF).  $\vec{R}_{jf}$  is the reaction forces at the hinge ‘jf’ produced by the sloshing mass whereas  $\vec{R}_{jp}$  is the reaction forces exchanged by the solar panel and the main platform at the fixed joint. Finally  $\vec{S}$  represents all the external forces (thrust, gravity and perturbations, if any) applied to the centre of mass of MP. The rotational equation of motion of the MP are the classical Euler equations of a rigid body that can be easily written as:

*Main platform rotational equations*

$$I_{cms} \dot{\vec{\omega}}_s + \vec{\omega}_s I_{cms} \vec{\omega}_s = \vec{d}_{jf} \vec{R}_{jf} + \vec{d}_{jp} \vec{R}_{jp} + \vec{M}_{jp} + \vec{C}_s + \vec{C}_{fs} \quad (4)$$

The terms  $I_{cms}$  and  $\vec{M}_{jp}$  represent the moment of inertia matrix of the MP and the reaction torque evaluated at the attachment point of the solar panel, respectively. Note that the symbol ( $\sim$ ) over the vectors  $\vec{\omega}_s$ ,  $\vec{d}_{jf}$  and  $\vec{d}_{jp}$  represents the skew-symmetric operator applied to these vectors. The right terms of Eq. (4) represent the total torque acting on the MP which is a combination of the effects produced by the reactions and by the external (gravity gradient, control and perturbations) torques here denoted with  $\vec{C}_s$  and by the motion of the fuel. For the case under concern (pendulum model with torsional damping) the torque exchanged between the rotation of the fuel and the tank of the satellite is expressed as [6]:

$$\vec{C}_{fs} = c(\vec{\omega}_f - \vec{\omega}_s) \quad (5)$$

Being  $\vec{\omega}_f$  and  $\vec{\omega}_s$  the angular velocity vector of the fuel and of the spacecraft respectively; and  $c$  the viscous damping coefficient of the fluid.

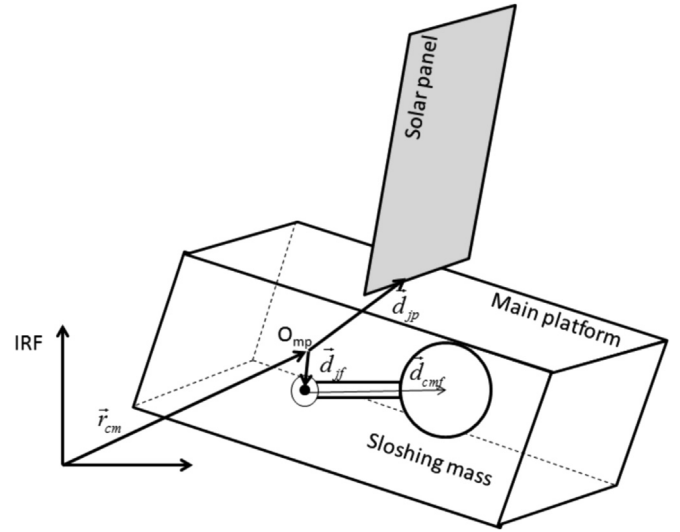


Fig. 1. Sketch of the floating multibody system.

Let us now consider the modelling of sloshing mass motion. As mentioned at the beginning of this section we decided to use a set of kinematic parameter which are already consistent with the compatibility conditions. On account of this, the position of the sloshing mass  $m_f$  with respect to the inertial reference frame can be written as follows:

$$\vec{r}_f = \vec{r}_{cm} + {}^I\mathbf{T}_s \vec{d}_{jf} + {}^I\mathbf{T}_f \vec{d}_{cmf} \quad (6)$$

where the matrices  ${}^I\mathbf{T}_s$  and  ${}^I\mathbf{T}_f$  allow to pass from the satellite and the sloshing mass reference frames to the inertial ones respectively. Here the vectors  $\vec{d}_{jf}$  and  $\vec{d}_{cmf}$  represent the position of the pivot of the spherical pendulum (in the spacecraft reference frame) and the position of the fuel sloshing mass with respect to the fuel reference frame respectively. By differentiating the Eq. (6) twice with respect to the time and by performing lengthy algebra, not reported here for the sake of brevity, we obtain the translational equilibrium of the fuel sloshing mass.

*Fuel slosh translational equations*

$$m_f \ddot{\vec{r}}_{cms} + m_f \vec{d}_{jf}^T \vec{\omega}_s + m_f \vec{\omega}_s \vec{\omega}_s \vec{d}_{jf} + m_f \vec{d}_{cmf}^T (\vec{\omega}_s + \vec{\omega}_f) + m_f \vec{\omega}_f \vec{\omega}_f \vec{d}_{cmf} = -\vec{R}_{jf} + \vec{F}_f \quad (7)$$

where  $\vec{\omega}_f$  is the angular velocity of the sloshing mass relative to the MP projected in the IRF,  $\vec{\omega}_f$  is the angular velocity of the sloshing mass with respect to the IRF and  $\vec{F}_f$  is the external force vector acting on the fuel sloshing. Finally the rotational equilibrium equation of the fuel slosh with respect to the pivot of the pendulum can be written as:

*Fuel slosh rotational equations*

$$m_f \vec{d}_{cmf} \ddot{\vec{r}}_{cms} + m_f \vec{d}_{cmf} \vec{d}_{jf}^T \vec{\omega}_s + m_f \vec{d}_{cmf} \vec{\omega}_s \vec{\omega}_s \vec{d}_{jf} + I_{jf} (\vec{\omega}_s + \vec{\omega}_f) + \vec{\omega}_f I_{jf} \vec{\omega}_f = \vec{C}_f + \vec{C}_{sf} \quad (8)$$

where  $\vec{C}_{sf} = -\vec{C}_{fs}$  and  $I_{jf}$  is the moment of inertia of the fuel mass as regards the pivot point. The vector  $\vec{C}_f$  is the torque produced by external forces acting on the fuel mass (if any).

The last body that constitutes the multibody spacecraft is the solar panel. For this body we must introduce the flexibility effects

that will be represented by the elastic displacements of the points of the solar array through the well-consolidated modal decomposition technique. On account of this the translational and the rotational equations of motion of the elastic panel (the macro-dynamics) will be coupled with the elastic (micro-dynamics) displacements represented by the amplitude of the modal shapes  $\eta_k$ . In general the coupling coefficients between flexible and rigid motion are represented by the so-called modal participation factors to the translation and to the rotation. Also in this case the lengthy exercise to derive the translation and rotation equations together with the generic mode equations for the solar panel is not repeated here for the sake of brevity (the mathematical details can be found in [16]):

#### Solar panel translational equations

$$\begin{aligned} m_p \ddot{l}_{cms} + m_p \ddot{d}_{jp}^T \ddot{\omega}_s + m_p \ddot{\omega}_s \ddot{\omega}_s \ddot{d}_{jp} + (m_p \ddot{d}_{cmp} + \ddot{\alpha} \ddot{\eta})^T \ddot{\omega}_s + \ddot{\omega}_s \ddot{\omega}_s (m_p \ddot{d}_{cmp} + \ddot{\alpha} \ddot{\eta}) \\ + 2 \ddot{\omega}_s \ddot{\alpha} \ddot{\eta} + \ddot{\alpha} \ddot{\eta} = - \ddot{R}_{jp} \end{aligned} \quad (9)$$

where  $\ddot{\alpha} = [\ddot{\alpha}_1 \dots \ddot{\alpha}_M]$  is the  $(3 \times M)$ -matrix containing the modal participation vectors  $\ddot{\alpha}_i$  (being  $M$  the number of selected modes),  $\ddot{\eta}$  is the  $(M \times 1)$ -vector of the modal amplitudes. Finally  $\ddot{d}_{cmp}$  is the vector that locates the position of the centre of mass of the solar panel in its undeformed configuration with respect to the position of its fixed joint on the MP (projected in the IRF). The operator  $(\circ)$  here is defined as  $\ddot{\alpha} \circ \ddot{\eta} = \sum_{i=1}^M \ddot{\alpha}_i \eta_i$ ; the term  $\ddot{\alpha}_i$  represents the skew-symmetric matrix of the vector  $\ddot{\alpha}_i$ .

#### Solar panel rotational equations

$$\begin{aligned} (m_p \ddot{d}_{cmp} + \ddot{\alpha} \circ \ddot{\eta})^T \ddot{l}_{cms} + (m_p \ddot{d}_{cmp} + \ddot{\alpha} \circ \ddot{\eta}) \ddot{d}_{jp}^T \ddot{\omega}_s \\ + (m_p \ddot{d}_{cmp} + \ddot{\alpha} \circ \ddot{\eta}) \ddot{\omega}_s \ddot{\omega}_s \ddot{d}_{jp} + I_{jp} \ddot{\omega}_s \\ + \ddot{\omega}_s I_{jp} \ddot{\omega}_s + (\ddot{\beta} \ddot{\eta}) = - M_{jp} \end{aligned} \quad (10)$$

where  $\ddot{\beta} = \ddot{\beta}_{3 \times M}$  is the rotational modal participation matrix and  $I_{jp}$  is the moment of inertia of the solar panel referring to the position of its fixed joint on the MP. Note that this matrix is in general not constant with respect to the time since it depends on the modal shapes and on the relevant modal amplitudes. However, such terms are very small if compared to the 'rigid' moment of inertia terms so that the matrix can be considered constant in practical cases.

#### Solar panel modal equations

In order to obtain the generic  $k$ -th mode equation we must project the solar panel 'local' equations on the  $k$ -th mode. The equation of equilibrium of a generic element of volume of the elastic body can be written as:

$$L[w] + \mu f_g + E_p - \mu a_p = 0 \quad (11)$$

where  $L[w]$  is the structural operator,  $\mu$  is the mass per unit volume,  $f_g$  is the vector of the gravitational force,  $E_p$  is the external force vector, other than gravitational, and  $a_p$  the acceleration of a point of the solar panel.

By applying the operation:

$$\int_B \varphi^T_k (L[w] + \mu f_g + E_p - \mu a_p) dB = 0 \quad (12)$$

being  $\varphi^T_k$  the transpose vector of the  $k$ -modal shape, we obtain the final set of modal equations (Eq. 13):

$$\begin{aligned} \ddot{\eta} + K \ddot{\eta} + \ddot{\beta}^T \ddot{\omega}_s + \ddot{\alpha}^T \ddot{l}_{cms} + \ddot{\alpha}^T \ddot{d}_{jp}^T \ddot{\omega}_s + \ddot{\alpha}^T \ddot{\omega}_s \ddot{\omega}_s \ddot{d}_{jp} \\ + \left\{ \begin{array}{c} \ddot{\omega}_s^T (I_{1,1} + I_{2,1})_1 \ddot{\omega}_s \\ \vdots \\ \ddot{\omega}_s^T (I_{1,M} + I_{2,M})_M \ddot{\omega}_s \end{array} \right\} + 2 \left[ \begin{array}{c} I_{3,1} \\ \vdots \\ I_{3,M} \end{array} \right] \ddot{\omega}_s = F_{\text{modal}} \end{aligned} \quad (13)$$

Here:

$$\begin{aligned} F_{1,k} &= \int \ddot{\varphi}_k \ddot{d}_p \mu dB, \\ F_{2,k} &= \sum_{j=1}^M \int \ddot{\varphi}_k \ddot{\varphi}_j \eta_j \mu dB, \\ F_{3,k} &= \sum_{j=1}^M \int \ddot{\varphi}^T_k \ddot{\varphi}^T_j \eta_j \mu dB, \end{aligned}$$

and  $F_{\text{modal}}$  is the vector containing the projection of the external forces on the elastic modes,  $K$  is the diagonal matrix of the square of the eigenfrequencies of the solar panel. Note that, since the panel is fixed to the MP, the projection of the reaction forces on the generic  $k$ -mode is null.

#### Assembled multibody governing equations

In the following the "assembled" governing equations of the multibody system are reported (Eq. 14).

$$\begin{aligned} \left[ \begin{array}{cccccc} m_s & 0 & 0 & 0 & -E & -E & 0 \\ 0 & l_{cms} & 0 & 0 & \ddot{d}_{jf}^T & \ddot{d}_{jp}^T & -E \\ m_f & m_f \ddot{d}_{jf}^T + m_f \ddot{d}_{cmf}^T & m_f \ddot{d}_{cmf}^T & 0 & E & 0 & 0 \\ m_f \ddot{d}_{cmf} & m_f \ddot{d}_{cmf} \ddot{d}_{jf}^T + I_{jf} & I_{jf} & 0 & 0 & 0 & 0 \\ m_p & m_p \ddot{d}_{jf}^T & 0 & \ddot{\alpha} & 0 & E & 0 \\ & + (m_{jp} \ddot{d}_{cmp} + \ddot{\alpha} \circ \ddot{\eta})^T & & & & & \\ (m_p \ddot{d}_{cmp} + \ddot{\alpha} \circ \ddot{\eta}) & (m_3 \ddot{d}_{cmp} + \ddot{\alpha} \circ \ddot{\eta}) \ddot{d}_{jp}^T & 0 & \ddot{\beta} & 0 & 0 & E \\ & + I_{jp} & & & & & \\ \ddot{\alpha}^T & \ddot{\alpha}^T \ddot{d}_{jp}^T + \ddot{\beta}^T & 0 & E & 0 & 0 & 0 \end{array} \right] \left\{ \begin{array}{c} \ddot{l}_{cms} \\ \ddot{\omega}_s \\ s \ddot{\omega}_f \\ \ddot{\eta} \\ R_{jf} \\ R_{jp} \\ M_{jp} \end{array} \right\} = \\ \left[ \begin{array}{c} 0 \\ -\ddot{\omega}_s l_{cms} \ddot{\omega}_s \\ -m_f \ddot{\omega}_s \ddot{\omega}_s \ddot{d}_{jf} - m_f \ddot{\omega}_f \ddot{\omega}_f \ddot{d}_{cmf} \\ -m_f \ddot{d}_{cmf} \ddot{\omega}_s \ddot{\omega}_s \ddot{d}_{jf} - \ddot{\omega}_f I_{jf} \ddot{\omega}_f \\ -m_p \ddot{\omega}_s \ddot{\omega}_s \ddot{d}_{jp} - \ddot{\omega}_s \ddot{\omega}_s (m_p \ddot{d}_{cmp} + \ddot{\alpha} \circ \ddot{\eta}) - 2 \ddot{\omega}_s \ddot{\alpha} \ddot{\eta} \\ -(m_p \ddot{d}_{cmp} + \ddot{\alpha} \circ \ddot{\eta}) \ddot{\omega}_s \ddot{\omega}_s \ddot{d}_{jp} - \ddot{\omega}_s (I_{jp}) \ddot{\omega}_s \\ -K \ddot{\eta} - \ddot{\alpha}^T \ddot{\omega}_s \ddot{\omega}_s \ddot{d}_{jp} - \left[ \begin{array}{c} \ddot{\omega}_s^T (I_{1,1} + I_{2,1})_1 \ddot{\omega}_s \\ \vdots \\ \ddot{\omega}_s^T (I_{1,M} + I_{2,M})_M \ddot{\omega}_s \end{array} \right] - 2 \left[ \begin{array}{c} I_{3,1} \\ \vdots \\ I_{3,M} \end{array} \right] \ddot{\omega}_s \end{array} \right] \left\{ \begin{array}{c} \ddot{S} \\ \ddot{C}_s + \ddot{C}_{js} \\ \ddot{F}_f \\ \ddot{C}_f + \ddot{C}_{sf} \\ F_p \\ C_p \\ F_1 \\ F_2 \\ \vdots \\ F_M \end{array} \right\} \quad (14)$$

Being  $E$  a  $(3 \times 3)$  unit matrix.

It is worth to note that the above system of equilibrium equations contains kinematic and kinetic (i.e. the reactions) unknowns. To solve this problem we can proceed through a back substitution by partitioning the system (14) in the following way:

$$\left[ \begin{array}{cc} \underbrace{A_{XX}}_{(9+M) \times (9+M)} & \underbrace{A_{XR}}_{(9+M) \times 9} \\ \underbrace{A_{RX}}_{(9) \times (9+M)} & \underbrace{A_{RR}}_{(9) \times (9)} \end{array} \right] \left\{ \begin{array}{c} \ddot{X} \\ R \end{array} \right\} = \left\{ \begin{array}{c} b_X \\ b_R \end{array} \right\} \quad (15)$$

From the second block of equations we can write:

$$[A_{RX}] \{ \ddot{X} \} + [A_{RR}] \{ R \} = \{ b_R \} \quad (16)$$

which is used to evaluate the generalized reaction force vector:

$$\{ R \} = [A_{RR}]^{-1} \{ b_R \} - [A_{RR}]^{-1} [A_{RX}] \{ \ddot{X} \} \quad (17)$$

The above vector can be now substituted into the first block leading to the final differential system that contains only the kinematics unknown:

$$[A_{XX} - A_{XR}A^{-1}_{RR}A_{RX}]\{\ddot{X}\} = \{b_X - A_{XR}A^{-1}_{RR}b_R\} \quad (18)$$

Note that the left-hand matrix practically coincides with the generalized mass matrix, which can be derived by employing the classical Lagrangian formulation. Of course the right hand side contains the non-linear terms, the elastic terms and also the external forces appearing in Eq. (14).

### 3. 2D mathematical model of a flexible spacecraft with fuel mass

In the present section the mathematical model of a 2D flexible spacecraft with a fuel sloshing mass will be reported. The general equations can be derived from the equations described in Section 2 providing that the multibody system is now two-dimensional and that the spherical pendulum can be represented as a revolute joint. Furthermore, as far as the elastic solar panel is concerned, the beam-like model is assumed in the present formulation. In Fig. 2 the schematic of the spacecraft is depicted. It consists of a Main Platform, a solar panel, here described as an elastic beam clamped to the spacecraft and of course by the sloshing mass. Of course not all the algebraic details are reported here for the sake of brevity. Suffice here to say that for the 2D case it is possible to perform the back substitution to evaluate the reaction (see Eqs. (15)–(17)) analytically. On account of this a complete set of non-linear second order ordinary differential equation is explicitly obtained as shown in Section 3.1.

#### 3.1. Dynamic equations for the 2D flexible spacecraft with a fuel sloshing mass

The set of kinematic variables here employed are the position vector of the centre of mass of the main platform ( $x_s, y_s$ ), the attitude angle  $\theta_s$  considering the x-axis of the ECI reference frame the angular position of the pendulum-like fuel slosh  $\theta_f$  and the amplitudes  $\eta_k$  ( $k = 1, M$ ) of the flexural elastic modes. Of course their derivatives with respect to the time will complete the set of the kinematic state variables.

Let us define, with  $m_s$ ,  $m_p$  and  $m_f$ , the masses of the main platform (plus the fixed mass of the fuel), of the solar panel and of the sloshing part of the fuel respectively. The position of the pivot of the pendulum with respect to the centre of mass of the MP is located along the longitudinal x-body axis of the MP and it is indicated with  $d_{jfx}$ ; the length of the pendulum, for this 2D case is indicated with  $d_{cmfx}$ ; finally the coordinates of the fixed-position of the solar panel with respect to the MP body axes is indicated by  $(d_{jpx}, d_{jpy})$ . The eigenmodes of vibration, normalized with respect to the mass of the beam-like solar panel, are in the following defined as  $\varphi_k$  and they are evaluated from the classical free-dynamic equations of the cantilever Euler–Bernoulli beam.

By employing the Eqs. (14)–(17) for the 2D case we obtain the final non-linear governing equations (Eq. (19)):

where  $I_j = I_f + m_f d_{cmfx}^2$  is the fuel slosh moment of inertia w.r.t the pivot point.

$$\begin{bmatrix} m_s + m_p + m_f & 0 & -m_f d_{jfx} \sin(\theta_s) - (m_p d_{jpy} + m_p d_{cmfx}) \cos(\theta_s) & -m_f d_{cmfx} \sin(\theta_f) & -\alpha_1 \cos(\theta_s) & \dots & -\alpha_M \cos(\theta_s) \\ & m_s + m_p + m_f & -(m_f d_{jfx} \cos(\theta_s)) - (m_p d_{jpy} + m_p d_{cmfx}) \sin(\theta_s) & m_f d_{cmfx} \cos(\theta_f) & -\alpha_1 \sin(\theta_s) & \dots & -\alpha_M \sin(\theta_s) \\ & & + (m_p d_{jpx} - \sum_k \eta_k \alpha_k \cos(\theta_s)) & & & & \\ & & I_{cms} + (m_f d_{jfx}^2) + (I_{jp} + m_p d_{cmfx} d_{jpy} + \sum_k m_k \eta_k^2) & m_f d_{jfx} d_{cmfx} & \beta_1 + d_{jpy} \alpha_1 & \dots & \beta_M + d_{jpy} \alpha_M \\ & & - \sum_k \alpha_k d_{jpx} + (d_{jpx} (m_p d_{jpx} - \sum_k \eta_k \alpha_k) & \cos(\theta_s - \theta_f) & & & \\ & & + (d_{jpy} (m_p d_{jpy} + m_p d_{cmfx})) & & & & \\ & \text{SYM} & & I_j & 0 & \dots & 0 \\ & & & & 1 & \dots & 0 \\ & & & & 0 & \ddots & \vdots \\ & & & & \vdots & \dots & 1 \end{bmatrix} \begin{Bmatrix} \ddot{x}_s \\ \ddot{y}_s \\ \ddot{\theta}_s \\ \ddot{\theta}_f \\ \ddot{\eta}_1 \\ \vdots \\ \ddot{\eta}_M \end{Bmatrix} = \begin{Bmatrix} \ddot{\theta}_s^2 [(m_f d_{jfx} + m_p d_{jpx} - \sum_k \eta_k \alpha_k) \cos(\theta_s) - (m_p d_{jpy} + m_p d_{cmfx}) \sin(\theta_s)] + \ddot{\theta}_f^2 [m_f d_{cmfx} \cos(\theta_f)] - 2\ddot{\theta}_s [\sum_k \eta_k \alpha_k \sin(\theta_s)] \\ \ddot{\theta}_s^2 [(m_f d_{jfx} + m_p d_{jpx} - \sum_k \eta_k \alpha_k) \sin(\theta_s) + (m_p d_{jpy} + m_p d_{cmfx}) \cos(\theta_s)] + \ddot{\theta}_f^2 [m_f d_{cmfx} \sin(\theta_f)] + 2\ddot{\theta}_s [\sum_k \eta_k \alpha_k \cos(\theta_s)] \\ - \ddot{\theta}_f^2 (m_f d_{jfx} d_{cmfx} \sin(\theta_s - \theta_f)) - 2\ddot{\theta}_s [\sum_k \eta_k (\eta_k - \alpha_k d_{jpx})] \\ \ddot{\theta}_s^2 (m_f d_{jfx} d_{cmfx} \sin(\theta_s - \theta_f)) \\ - \ddot{\theta}_s^2 (d_{jpx} \alpha_1 - \eta_1) - \omega_1^2 \eta_1 \\ \vdots \\ - \ddot{\theta}_s^2 (d_{jpx} \alpha_M - \eta_M) - \omega_M^2 \eta_M \end{Bmatrix} + \begin{Bmatrix} (F_E^f + F_g^f)_X + (F_g^s + F_g^f)_X \\ (F_E^f + F_g^f)_Y + (F_g^s + F_g^f)_Y \\ (F_E^f + F_g^f)_Y d_{jfx} \cos(\theta_s) - (F_E^f + F_g^f)_X d_{jfx} \sin(\theta_s) + G_g^s + C_E^s + T_{fs} \\ (F_E^f + F_g^f)_Y d_{cmfx} \cos(\theta_f) - (F_E^f + F_g^f)_X d_{cmfx} \sin(\theta_f) + C_d^f + T_{sf} \\ \int_0^{L_p} f_p(\xi) \varphi_1(\xi) d\xi \\ \vdots \\ \int_0^{L_p} f_p(\xi) \varphi_M(\xi) d\xi \end{Bmatrix} \quad (19)$$



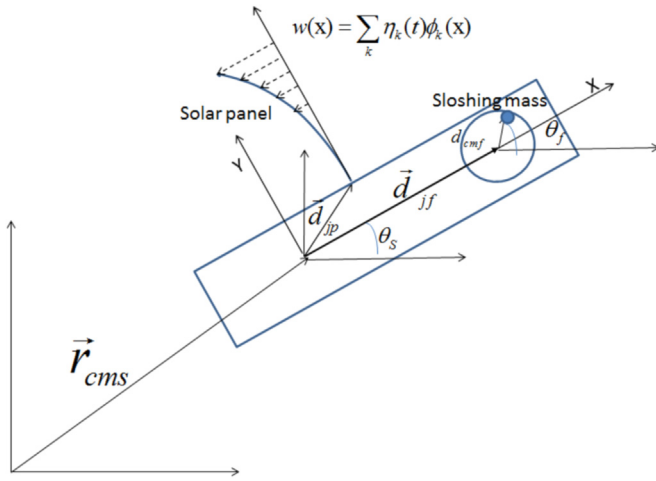


Fig. 2. 2D Space flexible spacecraft with fuel slosh.

Of course, even for this 'simplified' 2D model the governing equations appear complicated by the coupling and the non-linear terms.

### 3.2. Linearized equation for the 2D model

Let us suppose that:

- The spacecraft is in the initial state with:  
 $\theta_s, \dot{\theta}_s, \theta_f, \dot{\theta}_f, \eta_k, \dot{\eta}_k = 0, \theta_f = \theta_{if}$
- the thrust acting on the spacecraft is considered constant
- the initial state of angle of the sloshing mass with respect to the horizontal  $\theta_{if}$  will assume two different values  $(0, \pi)$  at the equilibrium.

Then the linear 2d-model can be described by the following equations:

Translation

$$\begin{aligned} M_{1,1}\ddot{x}_s + M_{1,3}\ddot{\theta}_s + M_{1,4}\ddot{\theta}_f + \sum_k M_{1,k}\ddot{\eta}_k &= F_x \\ M_{2,2}\ddot{y}_s + M_{2,3}\ddot{\theta}_s + M_{2,4}\ddot{\theta}_f + \sum_k M_{2,k}\ddot{\eta}_k &= F_y \end{aligned} \quad (20)$$

Rotation

$$\begin{aligned} M_{3,1}\ddot{x}_s + M_{3,2}\ddot{y}_s + M_{3,3}\ddot{\theta}_s + M_{3,4}\ddot{\theta}_f + \sum_k M_{3,k}\ddot{\eta}_k &= T_{orq,s} \\ M_{4,1}\ddot{x}_s + M_{4,2}\ddot{y}_s + M_{4,3}\ddot{\theta}_s + M_{4,4}\ddot{\theta}_f + \sum_k M_{4,k}\ddot{\eta}_k &= T_{orq,f} \end{aligned} \quad (21)$$

Mode equations

$$\begin{cases} M_{5,1}\ddot{x}_s + M_{5,2}\ddot{y}_s + M_{5,3}\ddot{\theta}_s + M_{5,4}\ddot{\theta}_f + M_{5,5}\ddot{\eta}_1 + \omega_1^2 \eta_1 \\ = F_{Mod,1} \\ M_{6,1}\ddot{x}_s + M_{6,2}\ddot{y}_s + M_{6,3}\ddot{\theta}_s + M_{6,4}\ddot{\theta}_f + M_{6,6}\ddot{\eta}_2 + \omega_2^2 \eta_2 \\ = F_{Mod,2} \\ \dots\dots\dots \\ M_{M+4,1}\ddot{x}_s + M_{M+4,2}\ddot{y}_s + M_{M+4,3}\ddot{\theta}_s + M_{M+4,4}\ddot{\theta}_f \\ + M_{M+4,M+4}\ddot{\eta}_M + \omega_M^2 \eta_M = F_{Mod,M} \end{cases} \quad (22)$$

Being:

$$M_{1,1} = M_{2,2} = m_s + m_p + m_f$$

$$M_{1,3} = -(m_p d_{jpy} + m_p d_{cmpx}) - (m_f d_{jfx}) \theta_s - m_p d_{jpx}$$

Table 1

Test cases analysed in the linear study.

| Control gains: $K_p, K_d$ | Fluid mass: $m_f$ (kg) | Thrust (N) | Relevant figures |
|---------------------------|------------------------|------------|------------------|
| 0, 0                      | 10                     | 1          | Fig. 3           |
| 0, 0                      | 10                     | 10         | Fig. 4           |
| 0.1, 0.01                 | 30                     | 1          | Fig. 5           |
| 1, 0.1                    | 10                     | 10         | Figs. 6 and 7    |
| 2, 0.2                    |                        |            |                  |
| 3, 0.3                    |                        |            |                  |
| 1, 0.1                    | 30                     | 10         | Fig. 8           |
| 2, 0.2                    |                        |            |                  |
| 3, 0.3                    |                        |            |                  |

$$M_{1,4} = M_{4,1} = -m_f d_{cmfx} \theta_f$$

$$M_{1,4+k} = M_{4+k,1} = -\alpha_k$$

$$M_{2,3} = -(m_f d_{jfx}) - (m_p d_{jpy} + m_p d_{cmpx}) \theta_s + (m_p d_{jpx})$$

$$M_{2,4} = M_{4,2} = -m_f d_{cmfx}$$

$$M_{2,4+k} = M_{4+k,2} = -\alpha_k \theta_s$$

$$\begin{aligned} M_{3,3} &= I_{cms} + (m_f d_{jfx}^2) + (I_{jp} + m_p d_{cmpx} d_{jpy}) \\ &+ (d_{jpx} (m_p d_{jpx}) + (d_{py} (m_p d_{jpy} + m_p d_{cmpx}))) M_{3,4} \\ &= m_f d_{jx} d_{cmfx} \cos(0 - \theta_{if}) \end{aligned}$$

$$M_{3,4+k} = M_{4+k,3} = \beta_k + d_{jpy} \alpha_k$$

$$M_{4,4} = I_j = I_f + m_f d_{cmfx}^2$$

$$M_{4,4+k} = M_{4+k,4} = 0$$

$$M_{k+4,k+4} = 1, k = 1, \dots, M$$

with  $\beta_k = \int \varphi_k \mu dB$  and  $\alpha_k = \int \xi \varphi_k \mu dB$  the modal participation factors for the 2D case. As mentioned above the normalized modal shapes  $\varphi_k$  of the solar panel is assumed, in the present work, equal to the ones of a flexible beam rigidly clamped to the bus of the satellite. Note also that the right terms appearing in (Eqs. (20)–(22)) represent the generalized forces acting on the centre of mass of the main platform, on the fluid and on the natural modes. Let us now use the Eq. (20) to explicitly express the components of the acceleration of the satellite as a function of the external forces and of the coupling inertial forces (i.e. rotation of the bus, rotation of the fluid and modal amplitudes):

$$\begin{aligned} \ddot{x}_s &= -\frac{M_{1,3}}{M_{1,1}} \ddot{\theta}_s - \frac{M_{1,4}}{M_{1,1}} \ddot{\theta}_f - \sum_k \frac{M_{1,k}}{M_{1,1}} \ddot{\eta}_k + \frac{1}{M_{1,1}} F_x \\ \ddot{y}_s &= -\frac{M_{2,3}}{M_{1,1}} + \ddot{\theta}_s - \frac{M_{2,4}}{M_{1,1}} \ddot{\theta}_f + \sum_k \frac{M_{2,k}}{M_{1,1}} \ddot{\eta}_k + \frac{1}{M_{1,1}} F_y \end{aligned} \quad (23)$$

Let now define:

$$M_{i,j}^* = \frac{M_{i,j}}{M_{1,1}} \quad (i = 1, M+2, j = 3, M+2)$$

and

$$F_x^* = \frac{F_x}{M_{1,1}}, F_y^* = \frac{F_y}{M_{1,1}},$$

By substituting Eq. (23) into Eq. (21) and into Eq. (22) we obtain the reduced version of the linearized form of the rotational and mode equations of the flexible spacecraft with the fuel sloshing mass:

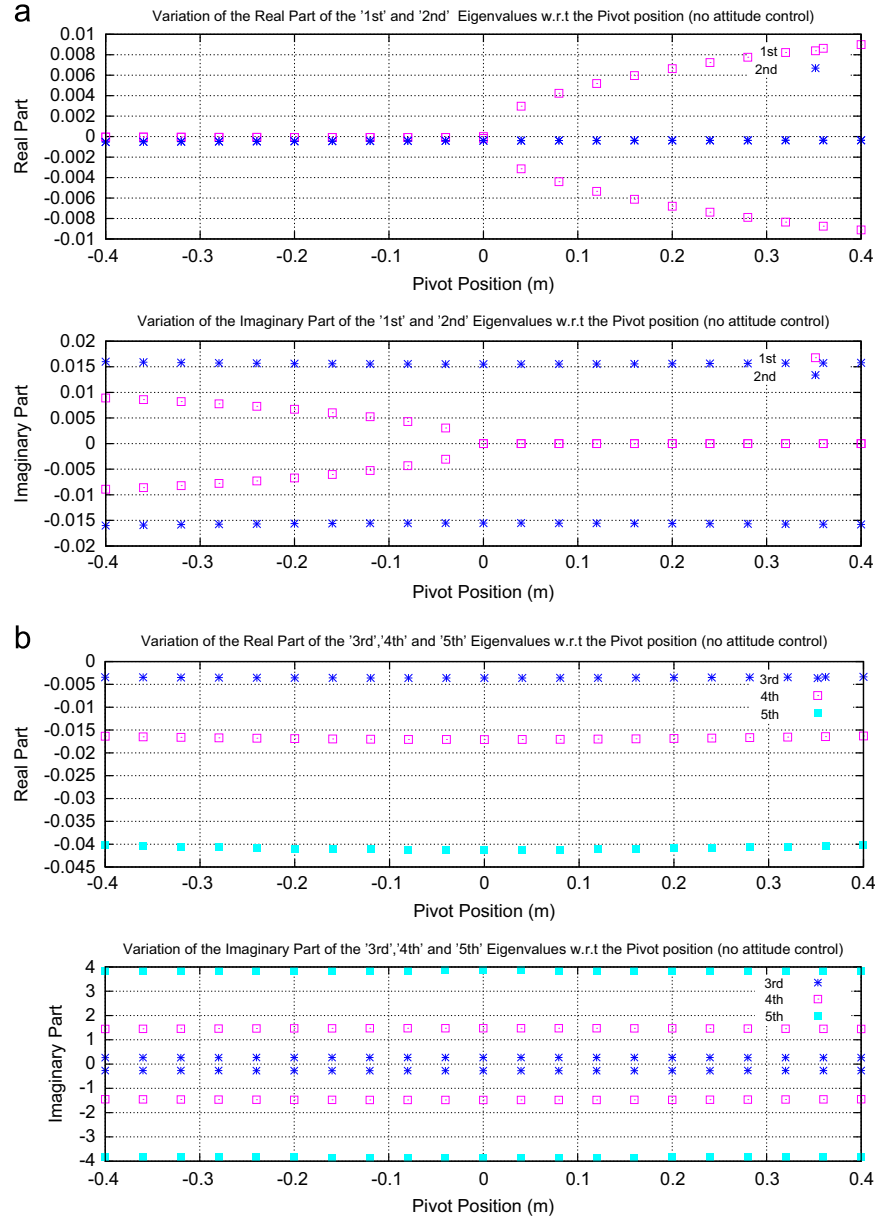


Fig. 3. Real and Imaginary parts of the eigenvalues w.r.t. the location of the pendulum (no attitude control,  $F_x^* = 8.66 \cdot 10^{-3}$  N/Kg).

#### Rotation

$$\begin{aligned} A_{1,1}\ddot{\theta}_s + A_{1,2}\ddot{\theta}_f + \sum_k A_{1,k}\ddot{\eta}_k &= T_{orq,s} - M_{3,1}F_x^* - M_{3,2}F_y^* = T_{\theta_s} \\ A_{2,1}\ddot{\theta}_s + A_{2,2}\ddot{\theta}_f + \sum_k A_{2,k}\ddot{\eta}_k &= T_{orq,f} - M_{4,1}F_x^* - M_{4,2}F_y^* = T_{\theta_f} \end{aligned} \quad (24)$$

#### Mode equations

$$\begin{cases} A_{3,1}\ddot{\theta}_s + A_{3,2}\ddot{\theta}_f + A_{3,3}\ddot{\eta}_1 + \omega_1^2 \eta_1 \\ \quad = F_{Mod,1} - M_{5,1}F_x^* - M_{5,2}F_y^* \\ A_{4,1}\ddot{\theta}_s + A_{4,2}\ddot{\theta}_f + A_{4,3}\ddot{\eta}_2 + \omega_2^2 \eta_2 \\ \quad = F_{Mod,2} - M_{6,1}F_x^* - M_{6,2}F_y^* \\ \dots \\ A_{M+2,3}\ddot{\theta}_s + A_{M+2,4}\ddot{\theta}_f + A_{M+2,M+2}\ddot{\eta}_M + \omega_M^2 \eta_M \\ \quad = F_{Mod,M} - M_{M+4,1}F_x^* - M_{M+4,2}F_y^* \end{cases} \quad (25)$$

Here the coefficients appearing in Eqs. (24) and (25) are defined as

follows:

$$A_{i-2,j-2} = M_{i,j} - M_{i,2}M_{2,j}^* - M_{i,1}M_{1,j}^* \quad (i = 3, \dots, M+2, j = 3, \dots, M+2)$$

The Eqs. (24) and (25) represent the final linearized equations of the system to be used to analyze the stability of the spacecraft providing that the right-hand terms are expressed as functions of state variables as we will see in the next session. Of course the set of the modal Eq. (25) can be complemented with the structural damping that can be easily added on the left-side terms by summing up the  $2\zeta_k\omega_k\dot{\eta}_k$  ( $k=1,M$ ). Here  $\zeta_k$  is the well-known damping ratio (conventionally taken  $=0.01$  for all the modes)

#### 4. Numerical examples

In this Section, some illustrative examples will be shown to study the dynamic behaviour of the spacecraft when a manoeuvre involving the thrust and the attitude control (for instance a station keeping manoeuvre) is considered. At first a parametric study to

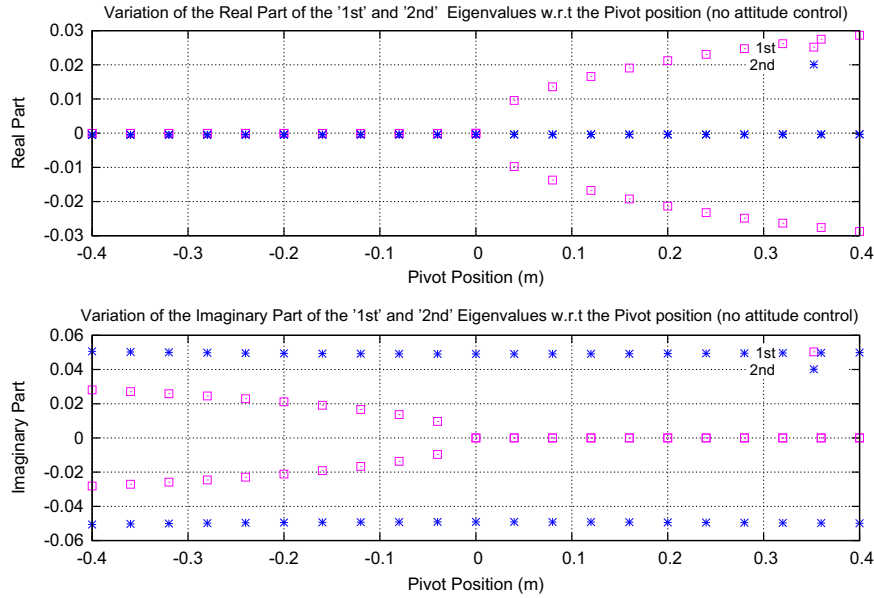


Fig. 4. Real and Imaginary parts of the eigenvalues w.r.t. the location of the pendulum  $d_{cmpx}$  (no attitude control,  $F_x^* = 8.66 \cdot 10^{-2}$  N/Kg).

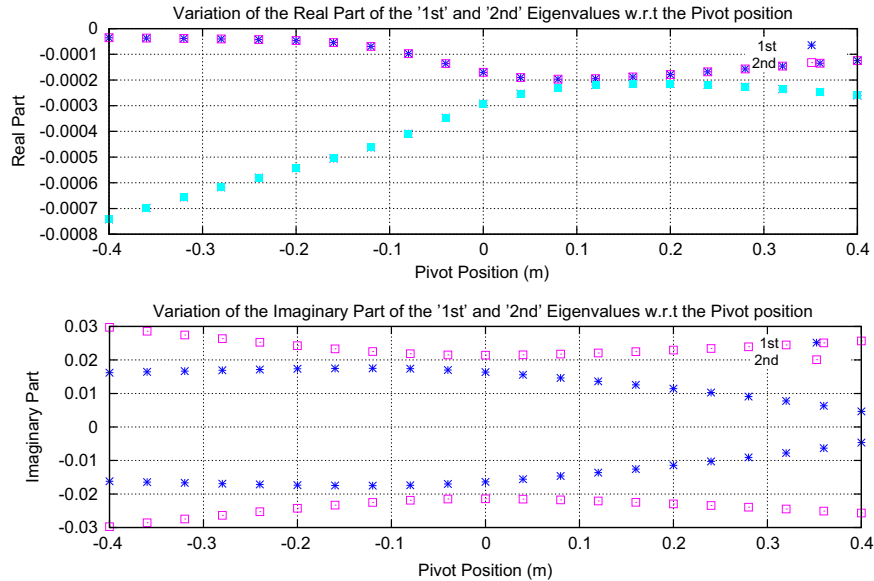


Fig. 5. Real and Imaginary parts of the eigenvalues w.r.t. the location of the pendulum  $d_{cmpx}$  ( $m_f = 30$  Kg,  $F_x^* = 7.38 \cdot 10^{-3}$  N/Kg, attitude control,  $k_{p\theta} = 0.1$ ,  $k_{d\theta} = 0.01$ ).

analyse the stability of the spacecraft for different values of the thrust, of the sloshing mass and of the gains of the attitude controller is reported in Section 4.1 (by considering the linearized model). Then in Section 4.2 the non-linear model is considered and several simulations are performed to evaluate the fully coupled behaviour of the space system under the action of complex manoeuvres.

#### 4.1. Coupling effects on the spacecraft dynamics

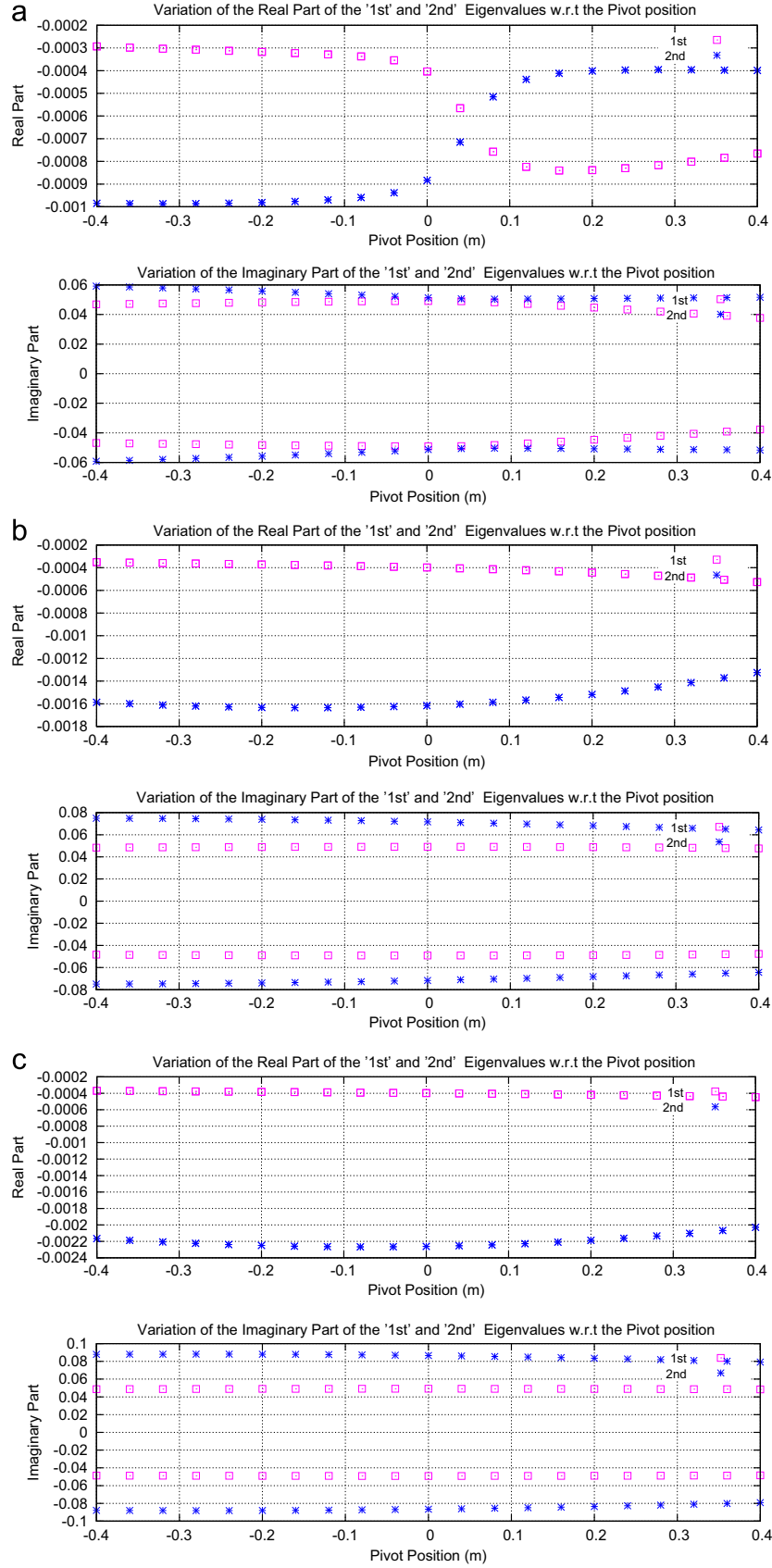
In order to perform a stability analysis on the linearized equations reported in Section 3.2 we have to express the right terms appearing in Eqs. (24) and (25) as function of the state variables. For the case under concern let us assume to neglect the terms connected to the gravity gradient forces since we are interested in analysing the effects produced by the thrust (that for the case under analysis is of orders of magnitude greater than the gravity gradient) and by the attitude control torque (if any).

By performing simple algebra, it is possible to observe that, at the equilibrium, the right terms appearing in Eq. (24) assume the following form:

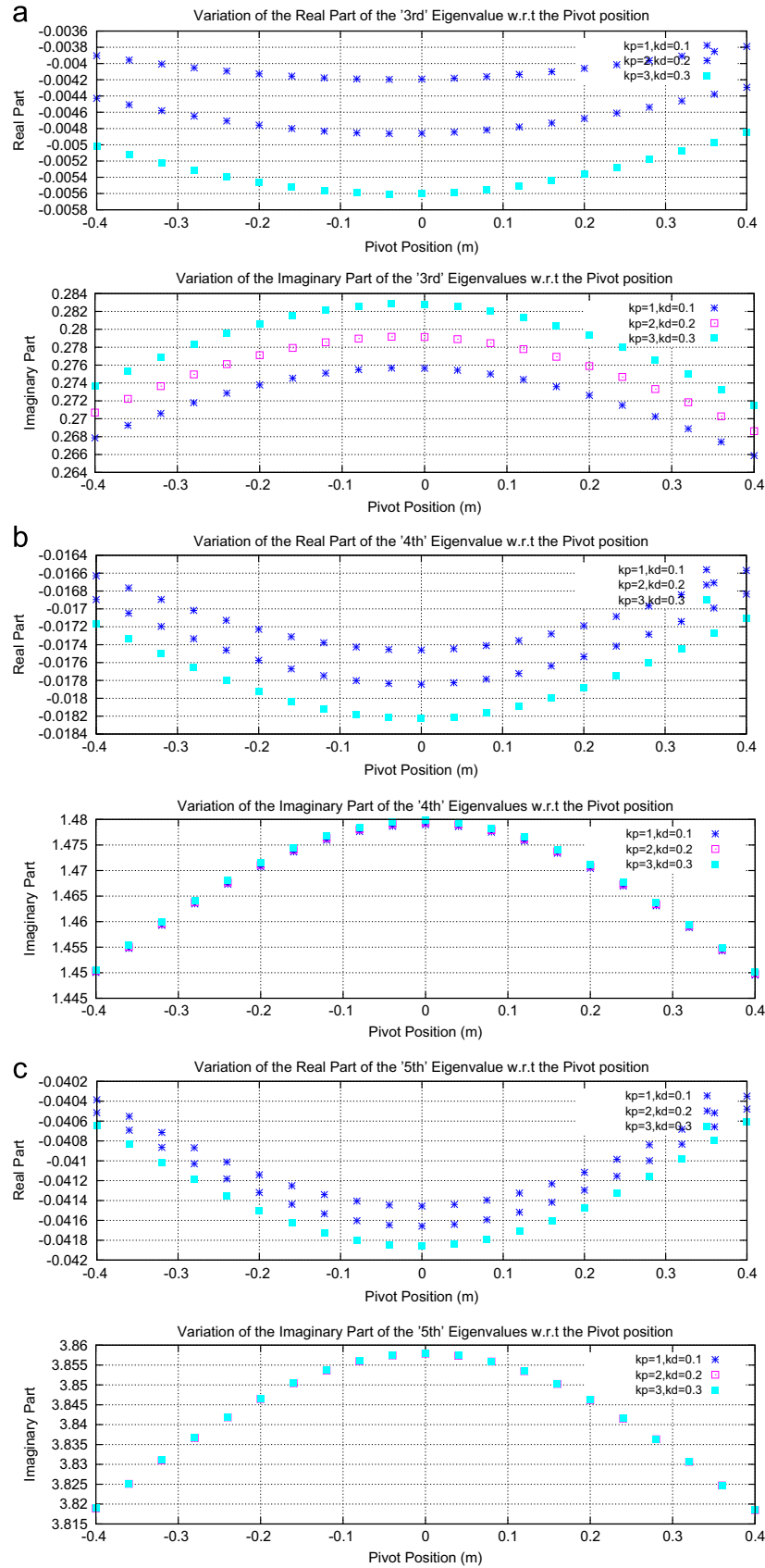
$$\begin{aligned} T_{\theta_s} &= m_f d_{jfx} \theta_s F_x^* + u_{\theta_s} + T_{js} \\ T_{\theta_f} &= m_f d_{cmfx} \cos(0 - \theta_{if}) F_x^* + T_{sf} \end{aligned} \quad (26)$$

Being  $F_x^*$  the module of the thrust along the x-axis,  $u_{\theta_s}$  the attitude control force acting on the spacecraft and  $T_{js} = -T_{sf} = c(\dot{\theta}_f - \dot{\theta}_s)$  the 2D version of Eq. (5). It is worth to note that, as far as the dynamics of the solar array is concerned (see Eq. (25)), we have three different terms on the right-hand side. The first one is the projection of the forces acting on the flexible panel on the modal shapes. These forces are practically null since: i) we have neglected the gravitational forces and ii) we have considered the panel clamped to the main bus so that the virtual work produced by the reaction forces at the fixed joint is null and the coupling effects between translational and rotational motion are

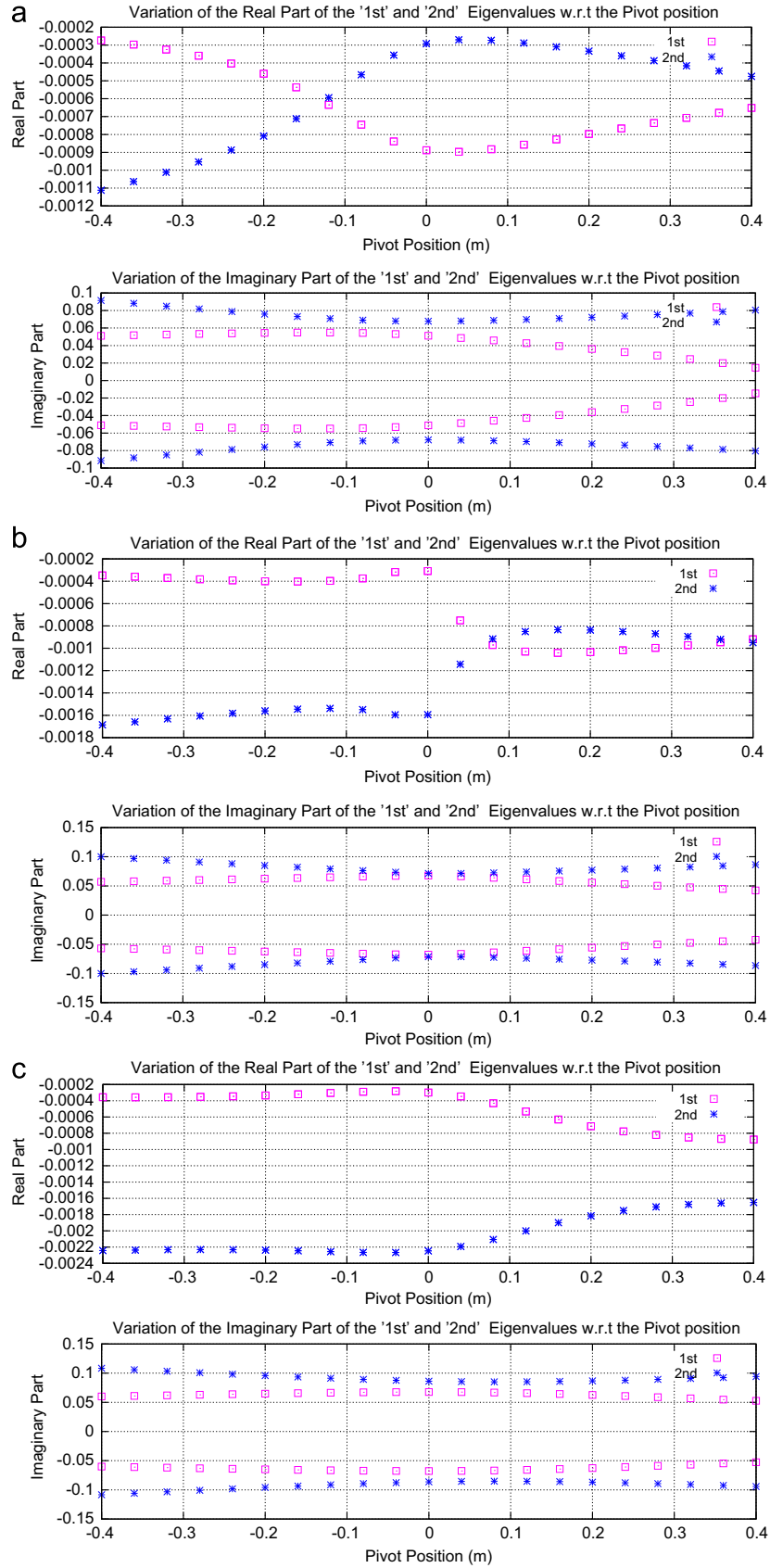




**Fig. 6.** Real and Imaginary parts of the eigenvalues w.r.t. the location of the pendulum  $d_{cmpx}$ , for different values of the attitude control gains ( $m_f = 10\text{Kg}$ ,  $F_x^* = 8.66 \cdot 10^{-2} \text{ N/Kg}$ ).



**Fig. 7.** Real and Imaginary parts of the eigenvalues associated to the dynamics of the solar panel w.r.t. the location of the pendulum  $d_{cmpx}$ , for different values of the attitude control gains ( $m_f = 10$  Kg,  $F_x^* = 8.66 \cdot 10^{-2}$  N/Kg).



**Fig. 8.** Real and Imaginary parts of the eigenvalues w.r.t. the location of the pendulum  $d_{cmpx}$ , for different values of the attitude control gains ( $m_f = 30$  Kg,  $F_x^* = 7.385510^{-2}$  N/Kg).

**Table 2**  
Test cases analysed in the nonlinear study.

| Reference attitude                                                               | Fluid mass          | Pivot position   | Relevant Figure |
|----------------------------------------------------------------------------------|---------------------|------------------|-----------------|
| $\vartheta_{rif} = 0^\circ$ (uncontrolled)                                       | $m_f = 10\text{Kg}$ | -0.4 m 0 m 0.4 m | Fig. 9          |
| $\vartheta_{rif} = 30^\circ$                                                     | $m_f = 10\text{Kg}$ | -0.4 m 0 m 0.4 m | Fig. 10         |
| $\vartheta_{rif} = 30^\circ \sin(\omega \cdot t) \omega = \frac{1}{20}\text{Hz}$ | $m_f = 10\text{Kg}$ | 0.4 m            | Fig. 11         |
| $\vartheta_{rif} = 30^\circ \sin(\omega \cdot t) \omega = \frac{1}{50}\text{Hz}$ | $m_f = 10\text{Kg}$ | 0.4 m            | Figs. 12 and 13 |
| $\vartheta_{rif} = 30^\circ \sin(\omega \cdot t) \omega = \frac{1}{20}\text{Hz}$ | $m_f = 30\text{Kg}$ | 0.4 m            | Figs. 14 and 15 |
| $\vartheta_{rif} = 30^\circ \sin(\omega \cdot t) \omega = \frac{1}{50}\text{Hz}$ | $m_f = 30\text{Kg}$ | 0.4 m            | Figs. 16 and 17 |

only due to the translational modal participation factors. The other two terms are connected to the thrust acting on the bus of the satellite. This thrust will produce a static (even if small) deformation on the panel as one can easily deduce by observing that the term  $M_{4+k,1} = -\alpha_k$  (which is practically the translational modal participation factor) that multiplies the external force  $F_x$  is not null. Of course, since we are interested now only in the evaluation of the stability of the dynamic system, we can remove this static force obtaining a full homogeneous linear ordinary differential system.

**4.1.1. Satellite characteristics.** The physical parameters used for the spacecraft model in this study are:

$$m_s = 100\text{Kg}, I_{cms} = 8.5 \text{ Kg m}^2, I_f = 1.5\text{Kg m}^2, m_f = 10\text{Kg}$$

The viscous damping necessary to model the friction between the fuel and the tank wall is assumed equal to  $c = 0.01 \text{ Nms}$ . The location of the pendulum base w.r.t the body fixed reference frame is  $d_{jfx} = 0.4 \text{ m}$ , the length of the pendulum is  $d_{cmfx} = 0.22 \text{ m}$ , the mass of the solar panel is  $m_p = 5.4 \text{ Kg}$  and its length is  $L_p = 2 \text{ m}$ . The position of the joint of the panel on the bus (in the bus reference frame) is considered  $d_{jpx} = 0$ ,  $d_{jpy} = 0.5 \text{ m}$ , the position of its centre of mass with respect to the joint is  $d_{cmplx} = 1 \text{ m}$ . In the present analysis the first three modal shapes of the solar panel are considered. On account of that the complete linearized satellite model has five degrees of freedom; two of them are associated to the attitude of the spacecraft and to the sloshing mass, the three remaining are associated to the modal amplitudes of the solar panel. The relevant natural frequencies are:  $[\omega_1 = 1.31, \omega_2 = 8.21, \omega_3 = 23.] \text{ rad/sec}$ .

The translational and rotational modal participation factors normalized with respect to the mass are respectively:

$$[\alpha_1 = -1.819, \alpha_2 = -1.008, \alpha_3 = -0.591] \sqrt{\text{kg}}$$

$$[\beta_1 = -2.643, \beta_2 = -0.421, \beta_3 = -0.150] (\text{m} \sqrt{\text{kg}})$$

The attitude control is considered a very simple proportional-derivative torque defined as:

$$u_{\vartheta_s} = -k_{p\vartheta}(\vartheta_s - \vartheta_{ref}) - k_{d\vartheta}\dot{\vartheta} \quad (27)$$

where of course the gains  $k_{p\vartheta}$  and  $k_{d\vartheta}$  must be opportunely chosen.

#### 4.2. Parametric stability analysis

First of all a parametric study is performed to analyse the effect of the position of the pendulum base (pivot position) with respect to the origin of the reference frame of the spacecraft on the natural frequencies of the damped multibody spacecraft.

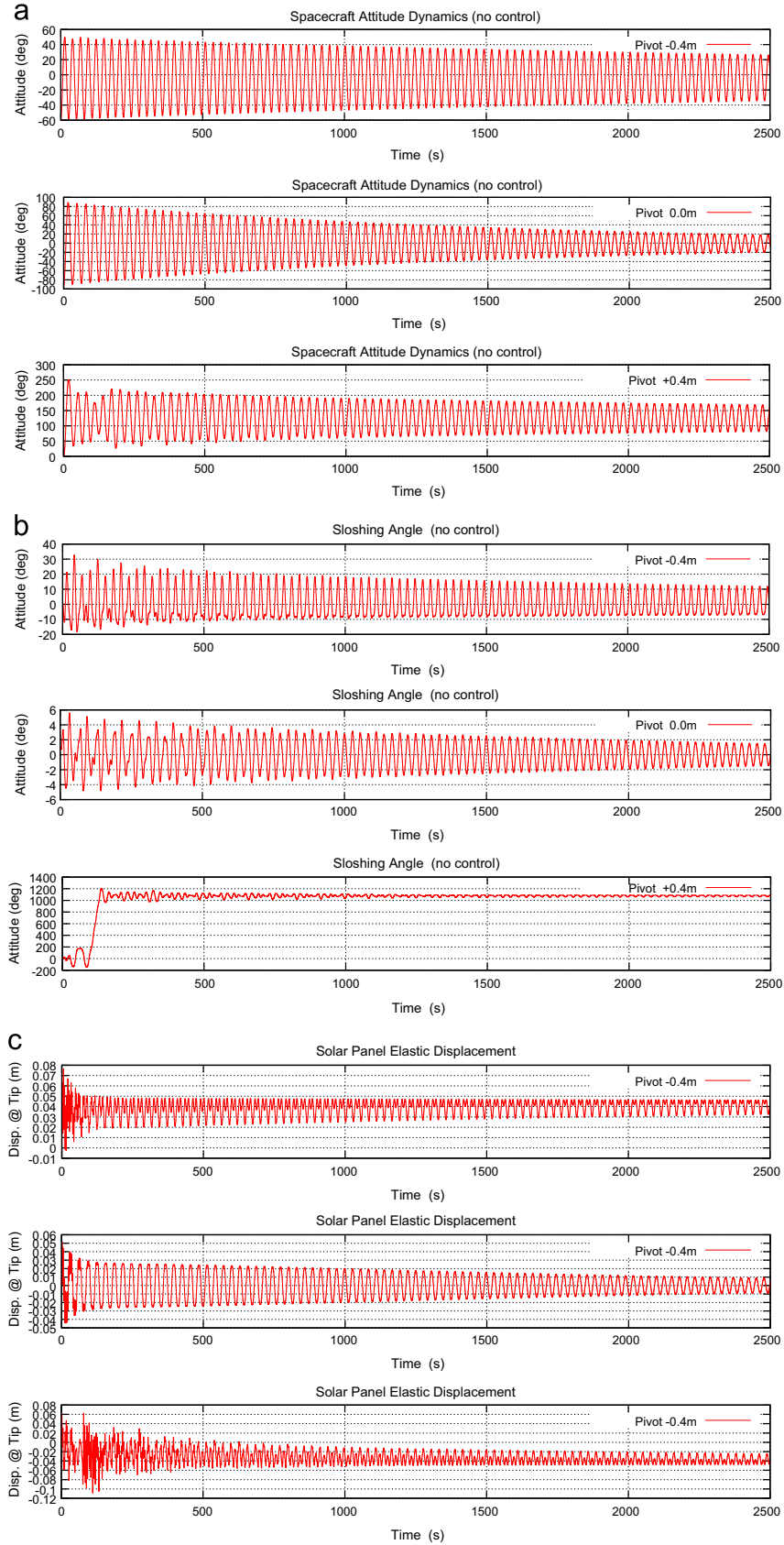
A number of parameters will be considered in this Section for understanding their influence on the linear stability of the system.

The simulations performed are summarized in Table 1.

As mentioned above there is one possible equilibrium position for the attitude of the spacecraft  $\vartheta_{sE} = 0$  (see Section 3.2) and two positions for the angle of the sloshing mass  $\vartheta_{fE} = 0, \vartheta_{fE} = \pi$ . It is possible to demonstrate that the case of  $\vartheta_{fE} = 0$  is an unstable configuration for the sloshing mass, meaning that the applied thrust will produce an un-stabilizing torque on the perturbed position the pendulum. More interesting is to analyse the behaviour of the spacecraft when the fluid mass is perturbed with respect to  $\vartheta_{fE} = \pi$ . In Fig. 3(a and b) the variation of the real and imaginary parts of the first five eigenvalues (the first two relevant to the attitude and the sloshing dynamics, the remaining relevant to the dynamics of the solar panel) when no attitude control is applied to the satellite are reported. It is worth noting that, even in the case of no control on the spacecraft, we observe a coupling among the elastic (the ones of the panel) and the 'rigid' degree of freedoms (i.e. the attitude and the sloshing mass).

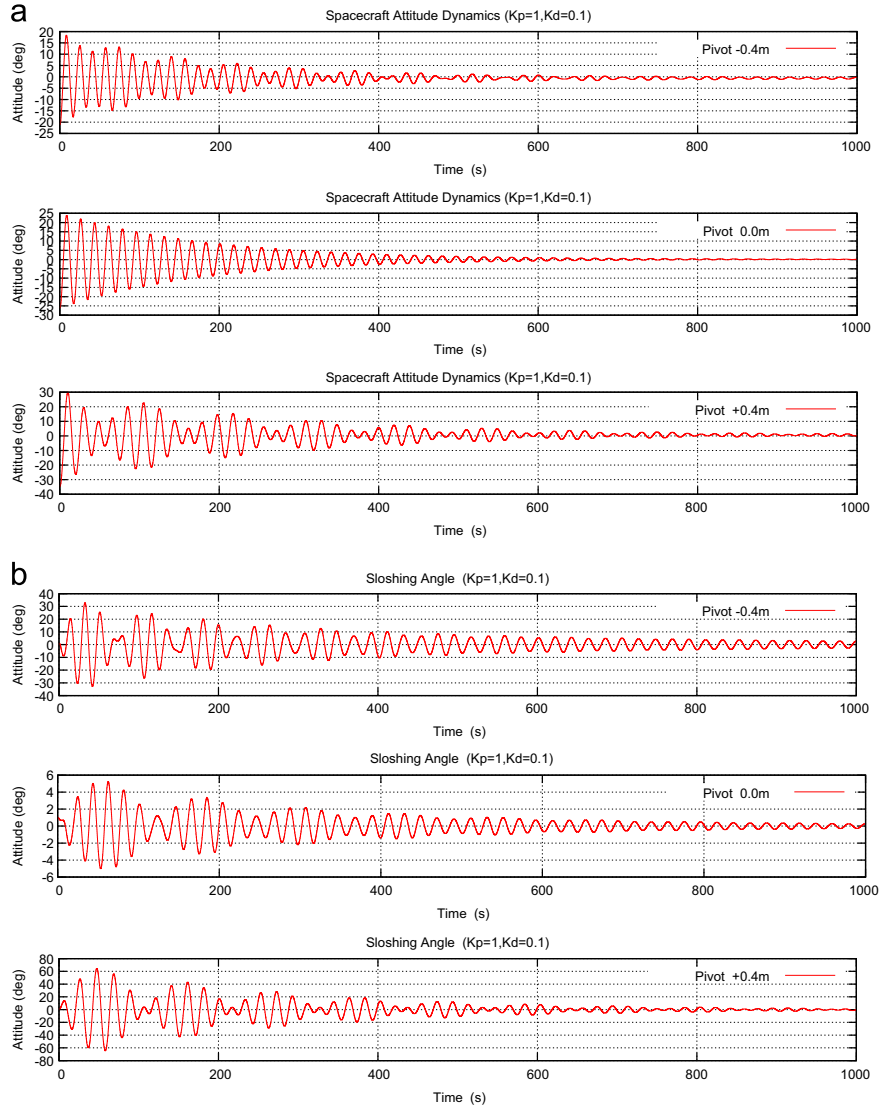
In fact, both imaginary parts of the first two eigenvalues are different from zero. It is also interesting to observe that one of the two eigenfrequencies (the one mainly associated to the fluid motion) remains almost constant with respect to the variation of the position of the pivot of the pendulum (see Fig. 3(a)). On the contrary, the imaginary part associated mainly to the attitude motion of the spacecraft is deeply altered by this position. This can be easily explained by analysing the term  $(m_f d_{jfx} F_x^*) \theta_5$  appearing in the first of Eq. (26) and hence on the right-hand of the first Eq. (24). It is evident that the thrust  $F_x^*$  (whose value is  $8.66 \cdot 10^{-3} \text{ N/Kg}$  equivalent to an applied force of 1 N) is coupled with the attitude dynamics of the spacecraft through the position of the pivot of the pendulum  $d_{jfx}$ . The final result is that an equivalent torsional spring is applied to the spacecraft whose value will determine a 'positive' or 'negative'-spring depending on the sign of the parameter  $d_{jfx}$ . A 'negative'-spring will result in a stabilizing effect on the attitude dynamics. Of course, the contrary is true when  $d_{jfx}$  is positive. The other three eigenvalues (reported in Fig. 3(b)) are associated to the damped normal modes of the flexible solar panel and they remain almost constant.

The results obtained when the value of the applied thrust is increased to 10 N, (i.e.  $F_x^* = 8.66 \cdot 10^{-2} \text{ N/Kg}$ ) are quite similar to the case of 1 N. We can observe, of course, an increase on the values of the natural frequencies associated to the attitude and to the sloshing mass motion (see Fig. 4) whereas the natural frequencies associated to the solar panel, not reported here for the sake of brevity, remain almost unchanged once again. Let us now consider the case of a sloshing mass  $m_f = 30\text{Kg}$ , which is almost the 30% of the mass of the plain spacecraft, when a PD attitude control ( $k_{p\vartheta} = 0.1, k_{d\vartheta} = 0.01$ ) is also considered in the linearized model. Fig. 5 shows an interesting phenomenon, common in several dynamic problems [17], where it seems that the imaginary part of



**Fig. 9.** Time histories of the attitude, the sloshing and of the elastic displacement of the solar panel for different position of the pendulum  $d_{cmpx}$ , ( $m_f = 10\text{Kg}$ ,  $F_x^* = 8.66 \cdot 10^{-2} \text{N/Kg}$ ), No Control.





**Fig. 10.** Time histories of the attitude and of the sloshing angle for different positions of the pendulum  $d_{cmpx}$ , ( $m_f = 10\text{Kg}$ ,  $F_X^* = 8.66 \cdot 10^{-2} \text{N/Kg}$ ), PD-Control.

the first two eigenvalues intersects each other. There is no intersection, but the two curves almost touch and then re-diverge. A similar behaviour is observed on the real parts which remain negative (i.e. the system is stable). As far as the natural frequencies of the solar panel (not reported here) are concerned, they remain practically the same.

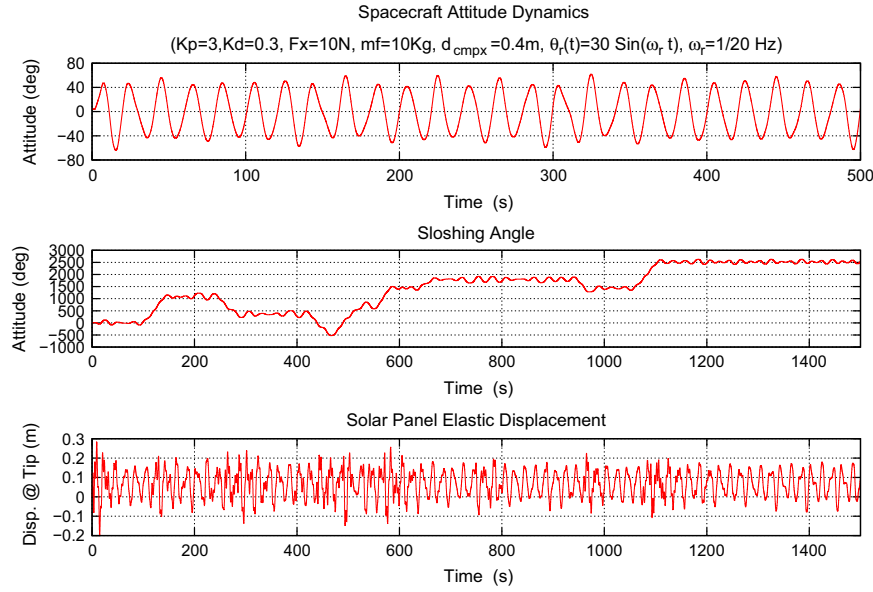
In order to better understand the coupling between the dynamic of the fluid mass and the attitude of the spacecraft when the attitude controller is taken into account other parametric analyses have been performed. In Fig. 6(a,b,c) the variation of the first two eigenvalues (real and imaginary parts) for different values of the gains are reported. Fig. 6(a) is referred to the case  $k_{p\theta} = 1, k_{d\theta} = 0.1$ , whereas Figs. 6(b) and (c) are referred to  $k_{p\theta} = 2, k_{d\theta} = 0.2$  and  $k_{p\theta} = 3, k_{d\theta} = 0.3$  respectively. It is interesting to observe that the phenomenon of the “coalescence” of the imaginary parts of the roots disappears as the attitude control gains increase. Furthermore, as one could expect, only the eigenvalue associated to the attitude dynamic is influenced by the variation of the gains. The one associated to the sloshing motion practically remains unchanged.

Another interesting effect can be observed on the real parts of the eigenvalues when  $k_{p\theta} = 1, k_{d\theta} = 0.1$  which are now strongly influenced by the position of the pivot (see the upper chart of the Fig. 6(a)) and they intersect each other. Finally in Fig. 7(a–c) the third, the fourth and the fifth eigenvalues (real and imaginary part) associated to the dynamics of the solar panel are reported. Note that the coupling of the elastic equations with the attitude and sloshing motion produces a slight variation on the values of the natural frequencies and of the damping of the solar panel.

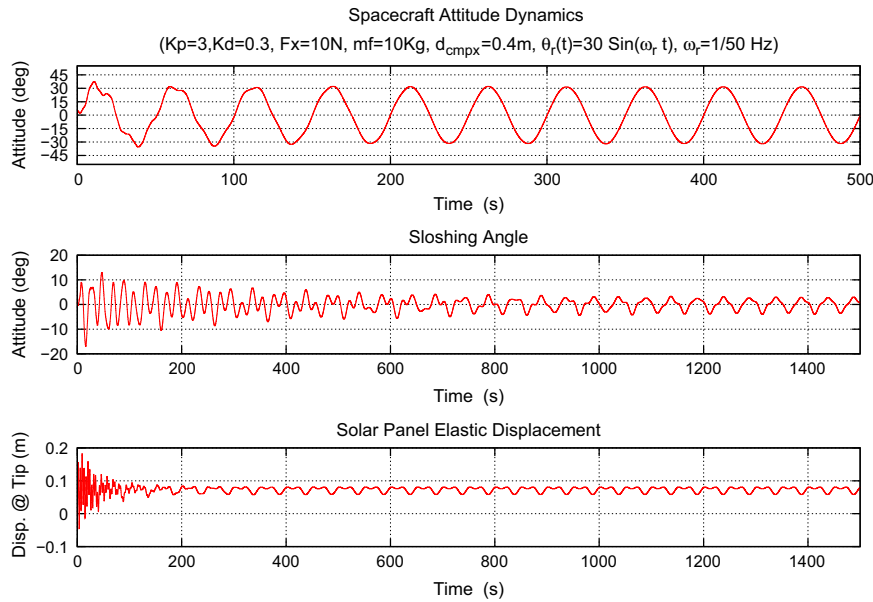
Similar results are obtained when the sloshing mass is increased to  $m_f = 30 \text{ Kg}$  as shown in Fig. 8(a–c). In this case the influence of the sloshing mass on the variation of the real part of the eigenvalues is much more evident with respect to the case of  $m_f = 10\text{Kg}$ .

#### 4.3. Time history analysis and control performance on the full non-linear model

In the present Section some numerical results based on the full non-linear model will be presented and discussed. Of course, for the non-linear case we are interested in “capturing” possible new phenomena not described by the linear model, if any.



**Fig. 11.** Time histories of the attitude, the sloshing and of the elastic displacement of the solar panel for a periodic attitude manoeuvre with  $\omega = 1/20$  Hz, ( $m_f = 10$  Kg,  $F_x^* = 8.66 \cdot 10^{-2}$  N/Kg,  $d_{cmpx} = .4$  m,  $k_p = 3, k_d = 0.3$ ).



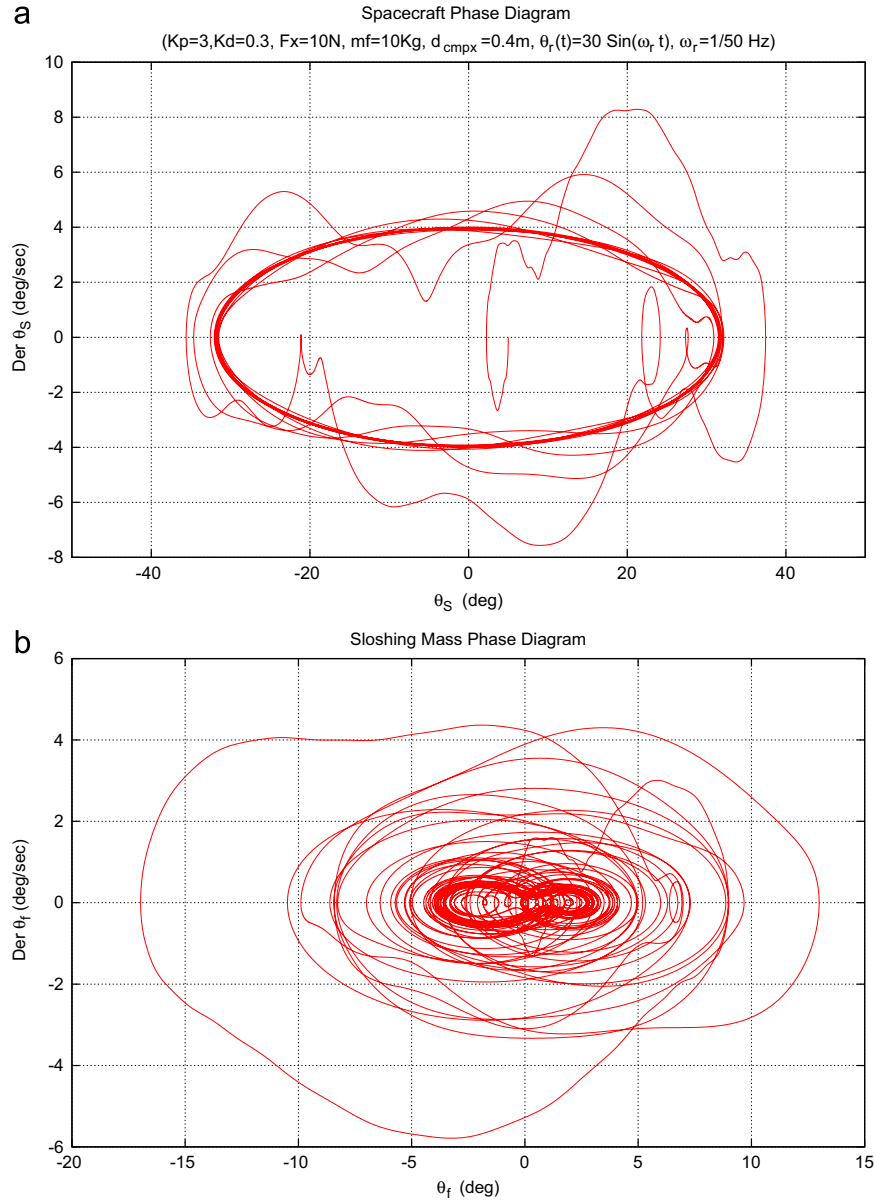
**Fig. 12.** Similar to Fig. 11 but with  $\omega = 1/50$  Hz.

The test cases considered investigate the effects of variations on the nonlinear dynamics of important parameters, such, fluid mass, the pivot position, the control gains, and finally the frequency of a time varying reference attitude. For the sake of clearness, in the following the Table 2 summarizes the test cases that will be analysed in this Section:

Fig. 9(a–c) shows the time histories of the attitude, of the angle of the sloshing mass and the elastic displacement measured at the tip of the solar array for three different values of the pivot (stable:  $d_{cmpx} \leq 0$ , unstable:  $d_{cmpx} > 0$  configurations) when an initial perturbation of  $\vartheta_s = 1^\circ$  is considered on the spacecraft (no attitude control). As far as the representation of the motion of the sloshing mass is concerned the angle ( $\vartheta_f(t) - \pi$ ) is reported to better represent the differences w.r.t the initial equilibrium position at  $\pi$ . The time histories reported on Fig.9 show supplementary

information with respect to the linearized parametric stability analysis. In fact, by observing Fig. 9(a), we see that although for  $d_{cmpx} \leq 0$  the attitude shows a periodic motion around the equilibrium position  $\vartheta_s = 0^\circ$  its amplitude reaches very high values. When  $d_{cmpx} > 0$  the satellite undergoes an attitude rotation greater than  $180^\circ$  in very few seconds and then it starts to rotate around a new equilibrium position of about  $130^\circ$ . Clearly this fact confirms the instability of the system (as also revealed by the linearized analysis when  $d_{cmpx} > 0$ ). Of course this behaviour has an impact on the motion of the fuel mass that starts to rotate around the stable initial condition and then moves to a new equilibrium configuration (see the bottom chart of the Fig. 9(b)) too. This unstable behaviour affects also the dynamics of the elastic panel as one can observe at the bottom of Fig. 9(c).

In order to evaluate the responsiveness of the PD attitude



**Fig. 13.** Phase space of attitude dynamics and sloshing motion,  $m_f = 10$  Kg,  $\omega = 1/50$  Hz.

control on the dynamics of the spacecraft, when the motion of the fluid and the flexibility of the solar panel are taken into account, an attitude re-orientation manoeuvre of  $\theta_{rif} = 30^\circ$  is now applied to the spacecraft. The relevant numerical results are reported in Fig. 10 when the PD gains are set to  $k_p = 3$ ,  $k_d = 0.3$ . It is worth to note that the required manoeuvre succeeds in about 600 s for all the pivot configurations, although a residual sloshing motion remains un-damped (about  $6^\circ$  in the worst case, first subplot in Fig. 10 (b)).

This is, of course, due to the small value on the fluid viscosity, which does not allow the damping out of the residual relative motion between the fluid and the wall of the tank.

Let us now analyse the case where the spacecraft is commanded to follow a periodic attitude manoeuvre i.e.  $\theta_{rif} = 30^\circ \sin(\omega \cdot t)$  being  $\omega$  an assigned frequency. The idea is to evaluate the influence of the gains of the PD controller on the non-linear attitude and fuel mass sloshing motion coupled with flexibility when the imposed attitude motion is not constant with respect to the time. In particular, two frequencies of the required

time varying manoeuvre are analysed, namely  $\omega = 1/20$  Hz and  $\omega = 1/50$  Hz.

Fig. 11 shows the time histories of the attitude of the spacecraft, of the sloshing angle and of the elastic displacement when the frequency of the required time varying manoeuvre is fixed to  $\omega = 1/20$  Hz. From the numerical simulation, it is evident that the motion of the fuel mass, besides being linearly unstable, also shows the classical behaviour of random non-periodic and chaotic motion typical of non-linear systems.

By decreasing the value of the frequency to  $\omega = 1/50$  Hz this chaotic motion disappears as shown in Fig. 12. This is also confirmed by observing the attitude and the sloshing motion on the phase plane as reported in Fig. 13 (a and b). It is evident that both trajectories, after a transient period, wrap into themselves generating stable limit cycles.

A similar analysis was conducted when the sloshing mass was increased to  $m_f = 30$  Kg. Fig. 14 shows the time histories whereas Fig. 15(a and b) the relevant phase plane plot when the periodic commanded attitude has a frequency of  $\omega = 1/20$  Hz.

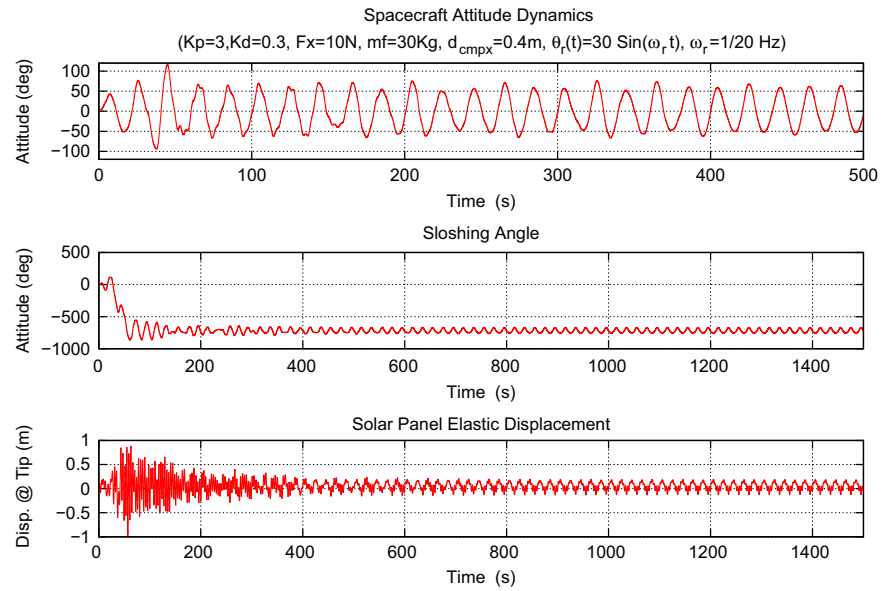


Fig. 14. Similar to Fig. 11 but with  $m_f = 30$  Kg.

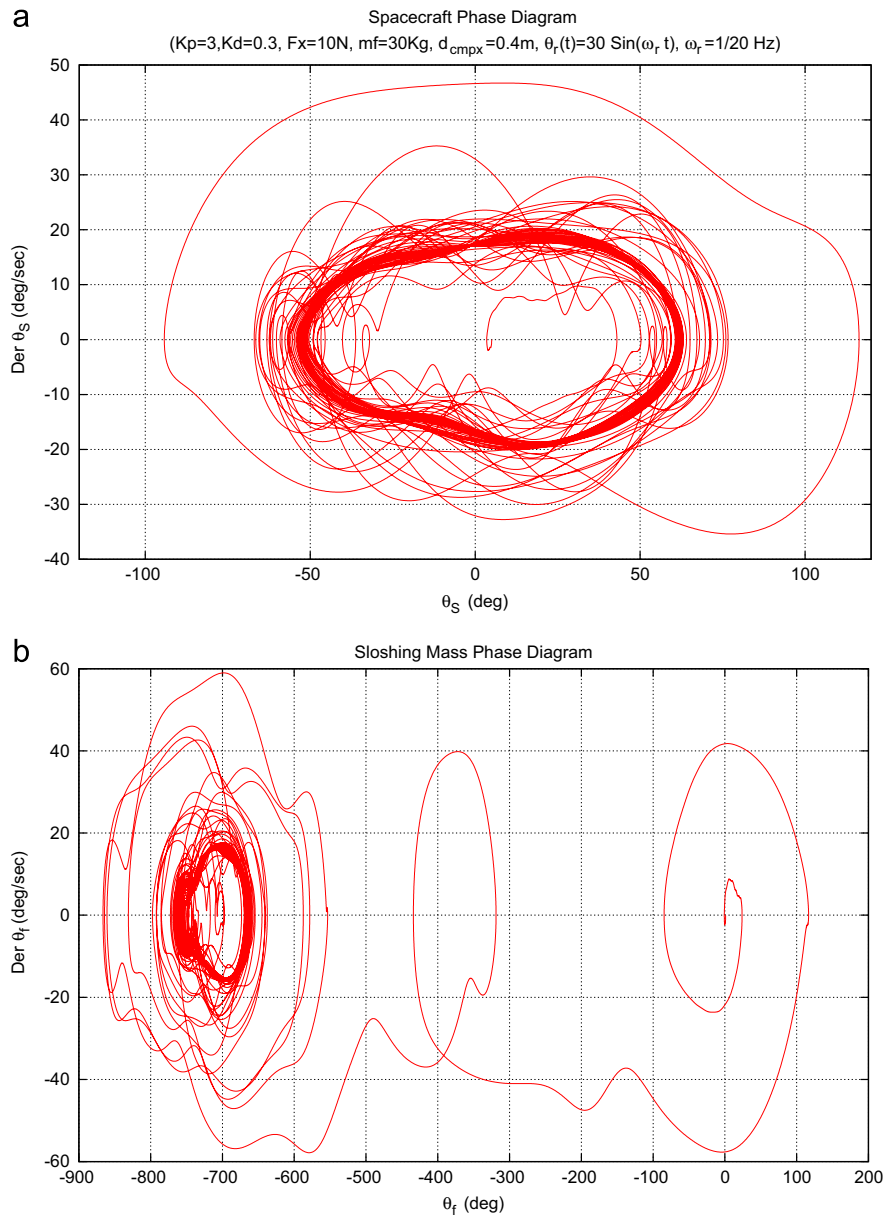
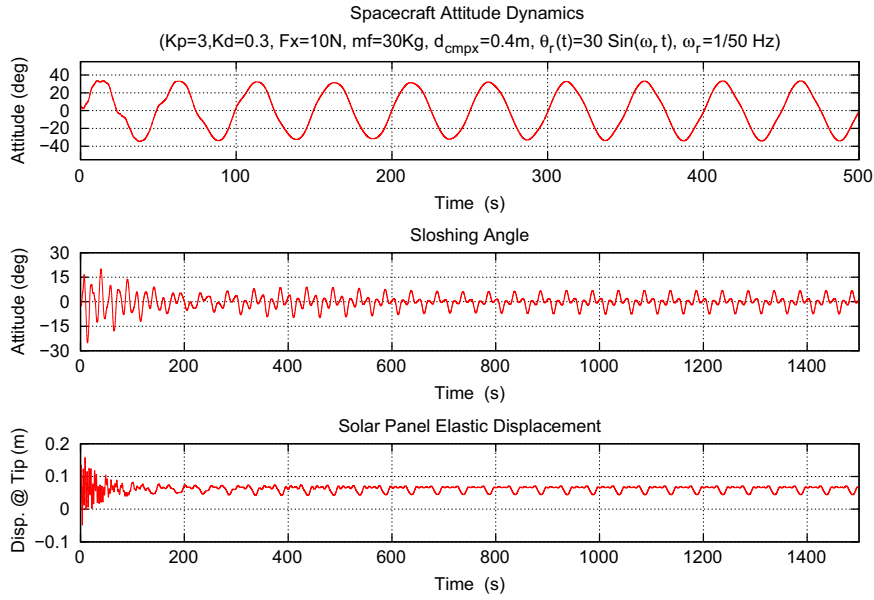
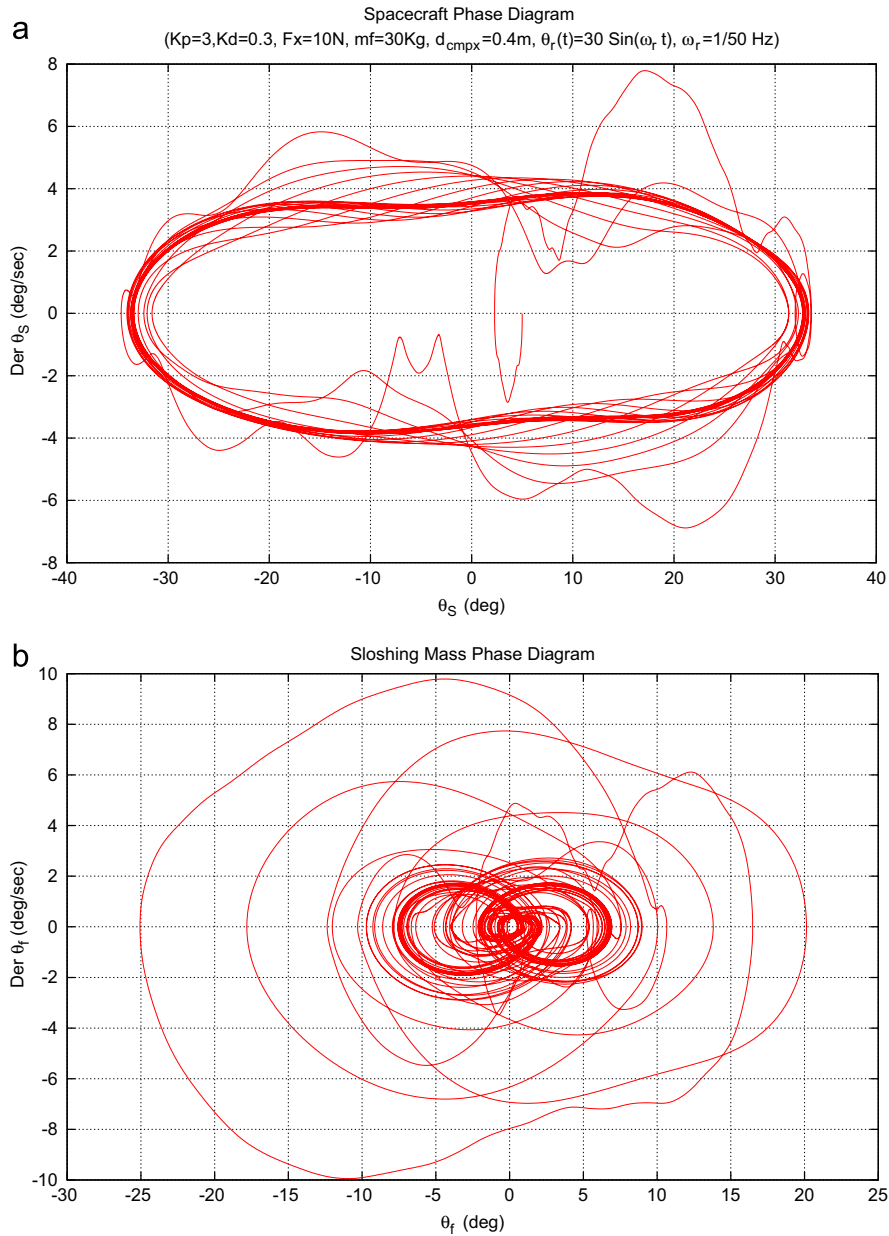


Fig. 15. Phase space of attitude dynamics and sloshing motion,  $m_f = 30$  Kg,  $\omega = 1/20$  Hz.



**Fig. 16.** Time histories of the attitude, the sloshing and of the elastic displacement of the solar panel for a periodic attitude manoeuvre with  $\omega = 1/50 \text{ Hz}$ , ( $m_f = 30 \text{ Kg}$ ,  $F_x^s = 8.66 \cdot 10^{-2} \text{ N/Kg}$ ,  $d_{\text{cmpr}} = 0.4 \text{ m}$ ,  $k_p = 3, k_d = 0.3$ ).



**Fig. 17.** Phase space of attitude dynamics and sloshing motion,  $m_f = 30 \text{ Kg}$ ,  $\omega = 1/50 \text{ Hz}$ .

Also in this case we observe that the ‘initial’ unstable motion of the sloshing mass becomes ‘stable’ around a complete rotated configuration as shown by the limit cycle reported in Fig. 15(b). Finally in Fig. 16 and in Fig. 17(a and b) a similar analysis is performed when  $\omega = 1/50$  Hz. Of course the increase of the fuel mass  $m_f$  is not only responsible of the instability of the dynamic motion of the pendulum but also on the performances of the PD attitude controller which is now able to regulate the attitude of the spacecraft to the desired path (see the upper charts of Figs. 16 and 17(a)).

## 5. Conclusions

The general formulation of a flexible spacecraft with a fuel sloshing mass described through a spherical pendulum model has been developed. Indeed due to the complexity of the space system the governing equations were obtained using a multi-body approach where the main bus, the sloshing mass and the flexible appendage attached to the rigid spacecraft are considered separated bodies connected to each other through joints and hence through the relevant reaction forces. To better understand the coupling effects among the rigid, flexible and fuel dynamics a “simplified” 2D model was derived from the full 3D formulation. The relevant equations are then used in two ways. In the first one a linearized version of the 2D equations is used to analyse the influence of some parameters (i.e. the fuel sloshing mass, the applied thrust and the position of the tank) on the stability of the system. Then, the fully non-linear equations were used to analyse the behaviour of the spacecraft when attitude manoeuvres are prescribed and the applied controller is a simple PD. The numerical results showed that the linearized system is able to catch only some aspects of the complex interaction among the fuel motion, the dynamics of the solar panel and the attitude of the spacecraft. Indeed non-linear dynamic model is required to fully describe the behaviour of the spacecraft under the sloshing motion. This aspect is very important when more sophisticated attitude controllers must be designed.

The present study should be considered as preparatory for

developing more complex control algorithms for an accurate attitude control even in presence of disturbing actions such as the elastic vibrations, coupled with the fuel sloshing.

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